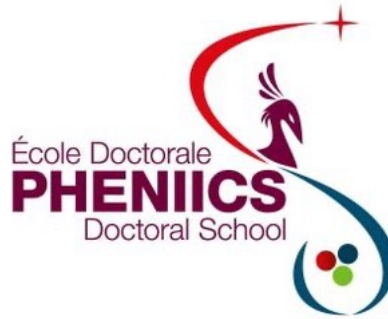




rgB Meeting

ϕ electroproduction on the neutron with CLAS12

22 May 2026

M. Ronayette, P. Chatagnon

ϕ electroproduction on the neutron with CLAS12

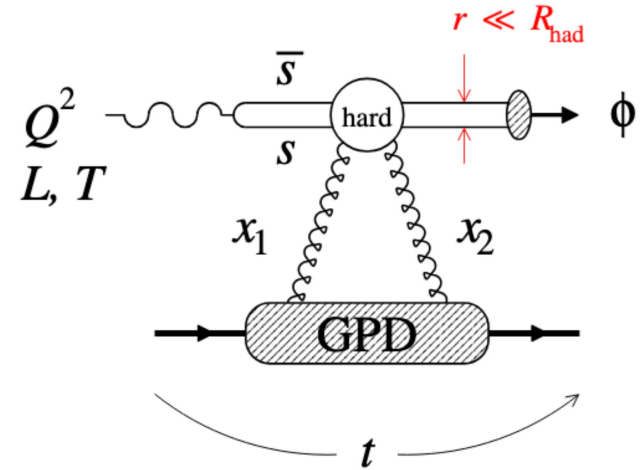
1. Motivations and events selection
2. Update on dataset used : spring2020 and spring2019
3. Update on the acceptance : kaons smearing
4. Update on the acceptance : new method
5. Update on the simulation : impact of the SDME

Motivations

Analysis objective :

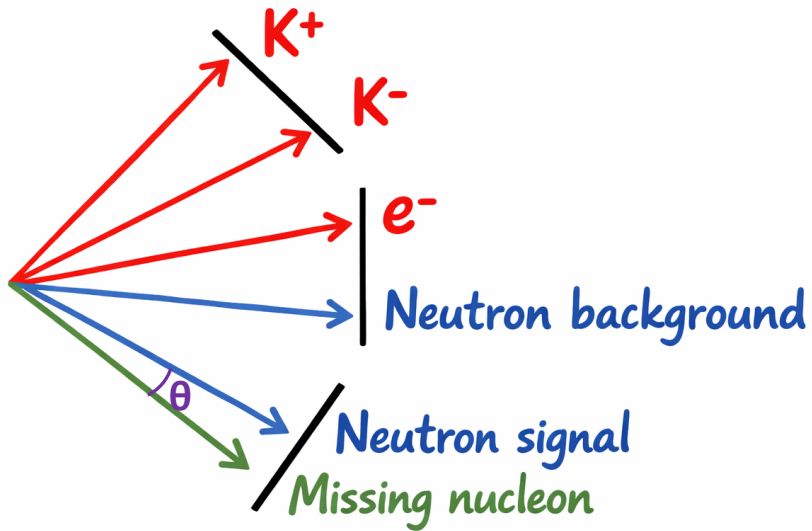
Measurement of the cross section of the ϕ electroproduction on **neutron** in the $K^+ K^-$ channel with **deuterium** target.

$e n \rightarrow e n' K^+ K^- \rightarrow$ Access to the **gluon** content of the neutron



Change of t -dependence with Q^2
 $\rightarrow$ gluonic radius of the neutron

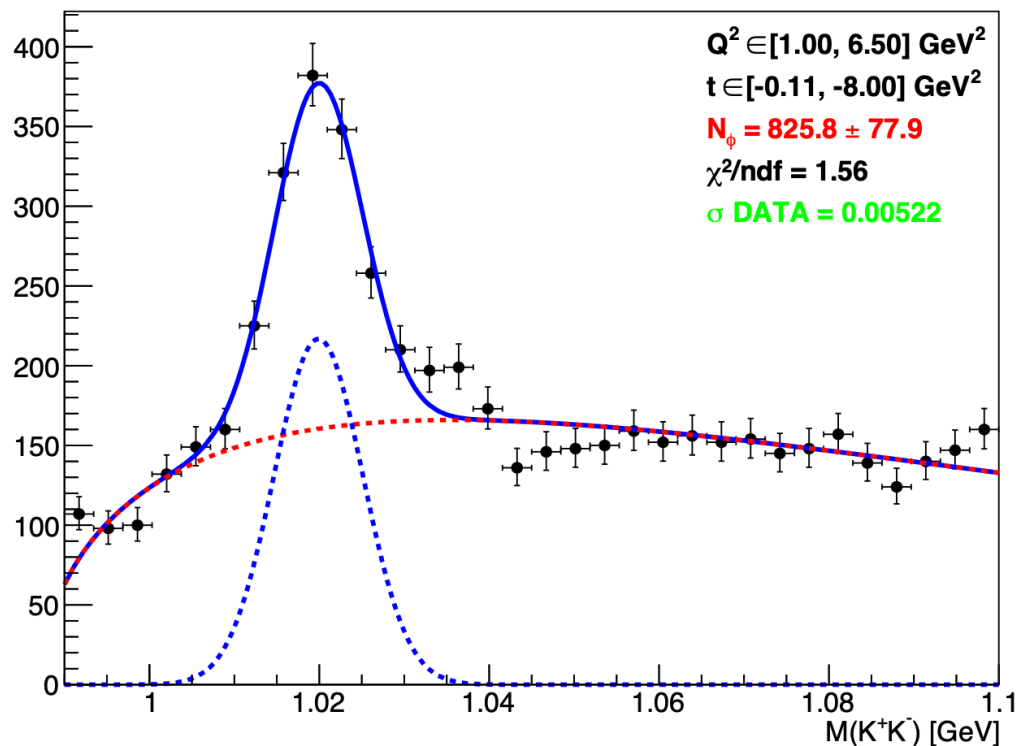
Events selection :



Select the best neutron which minimize the angle between the **missing nucleon** and the **neutron** and verify that the angle θ between **the missing nucleon** and the **neutron** is less than 5° . Also we need 0 proton detected to reduce contamination.

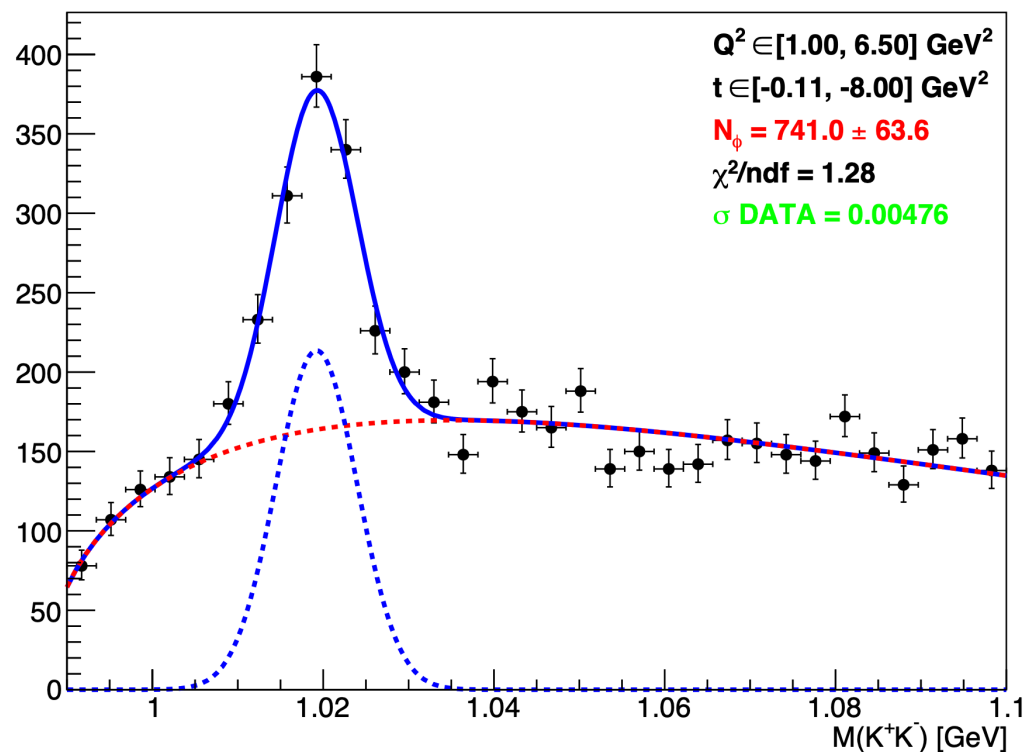


Update on dataset : look spring2020



Fall2019 outbending

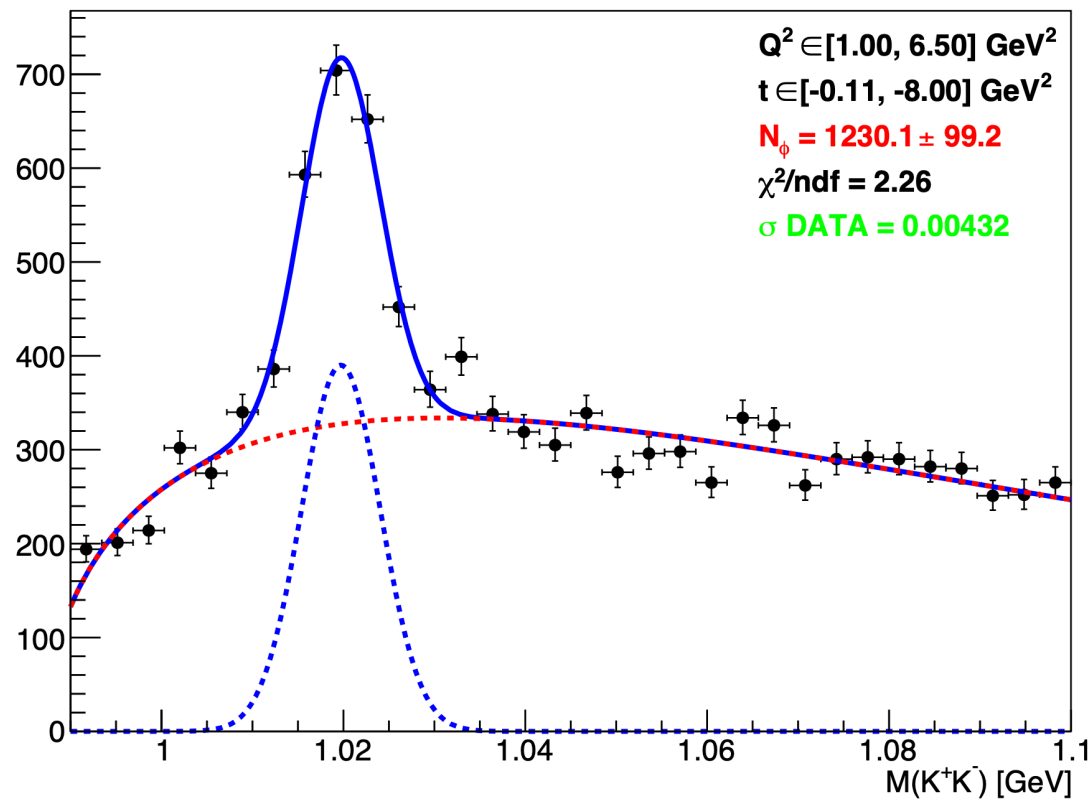
Q = 12.1 mC



Spring2020 inbending

Q = 28.3 mC

Update on dataset : look spring2019

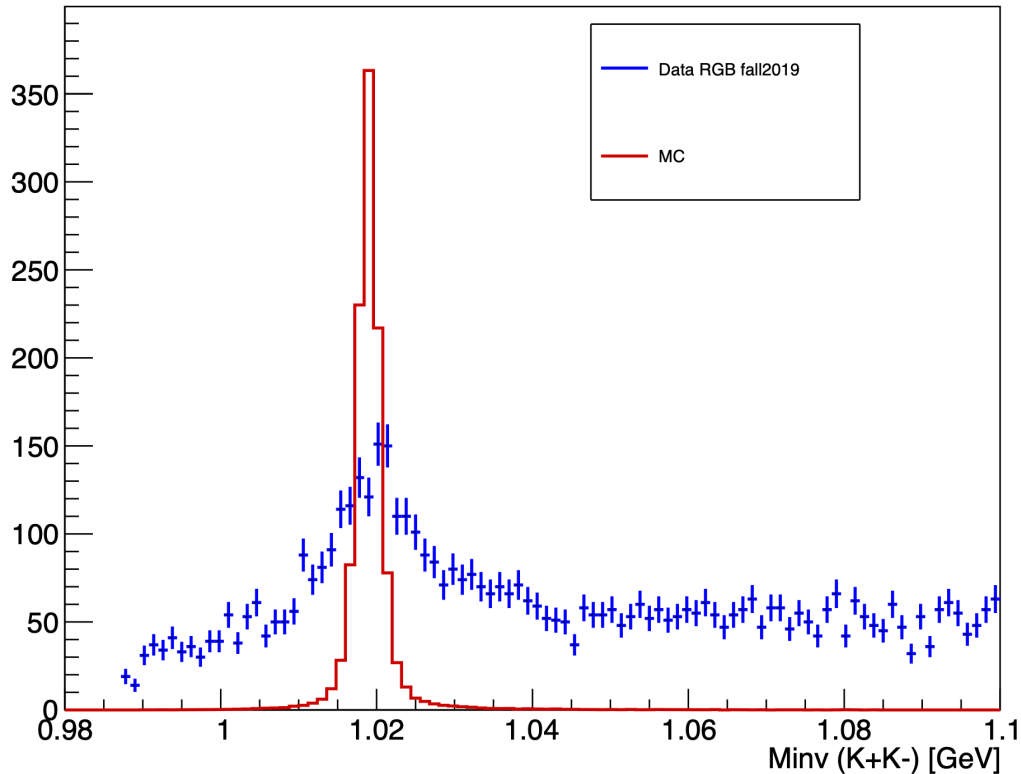


Spring 2019 inbending

Q = 63.4 mC

Smearing of the kaons

Invariant Mass K+ K-



→ We need a smearing of both kaons in the MC

$$P_{smear} = P \cdot (1 + \eta_s)$$

With η_s a random number from a Gaussian centered at 0.

The size of the Gaussian is the parameter $\delta(P)$

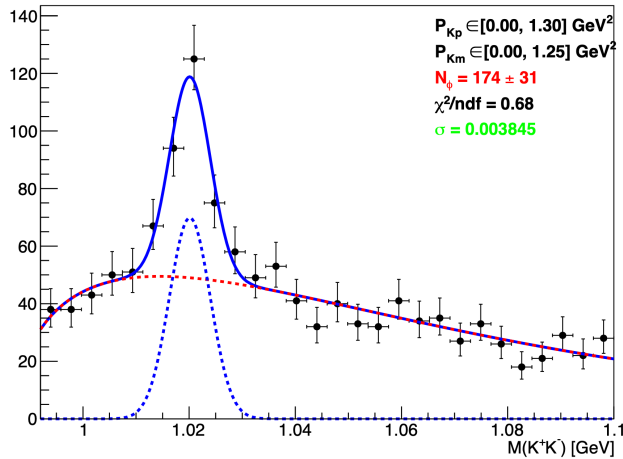
$$\delta_{K^+}(P) = A_1 \Theta(P - P_1) \Theta(P_2 - P) + B_1 \Theta(P - P_2) \Theta(P_3 - P) + \dots$$

$$\delta_{K^-}(P) = A_2 \Theta(P - P_1) \Theta(P_2 - P) + B_2 \Theta(P - P_2) \Theta(P_3 - P) + \dots$$

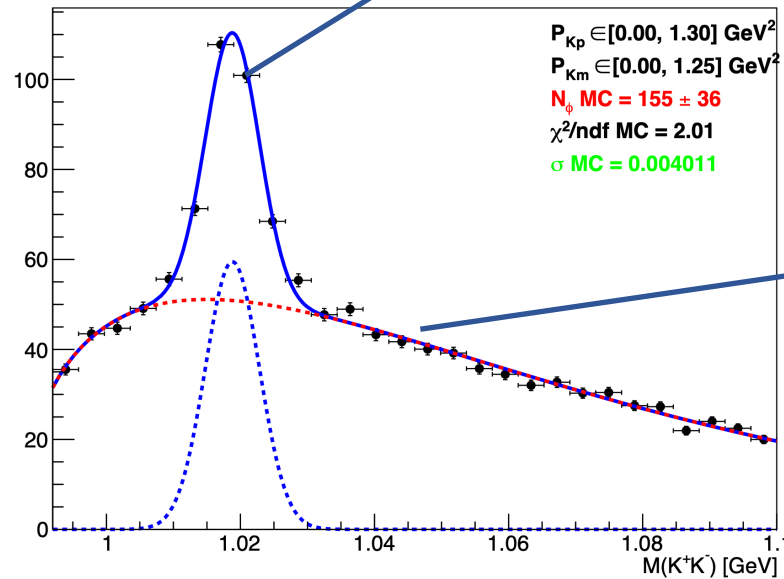
→ Minimize $\sigma_{\text{DATA}} - \sigma_{\text{MC}}$ per bins of momentum to find the parameters $A_1, B_1, C_1, \dots$

Smearing of the kaons

How to extract σ_{MC} in one bin of momentum?



DATA



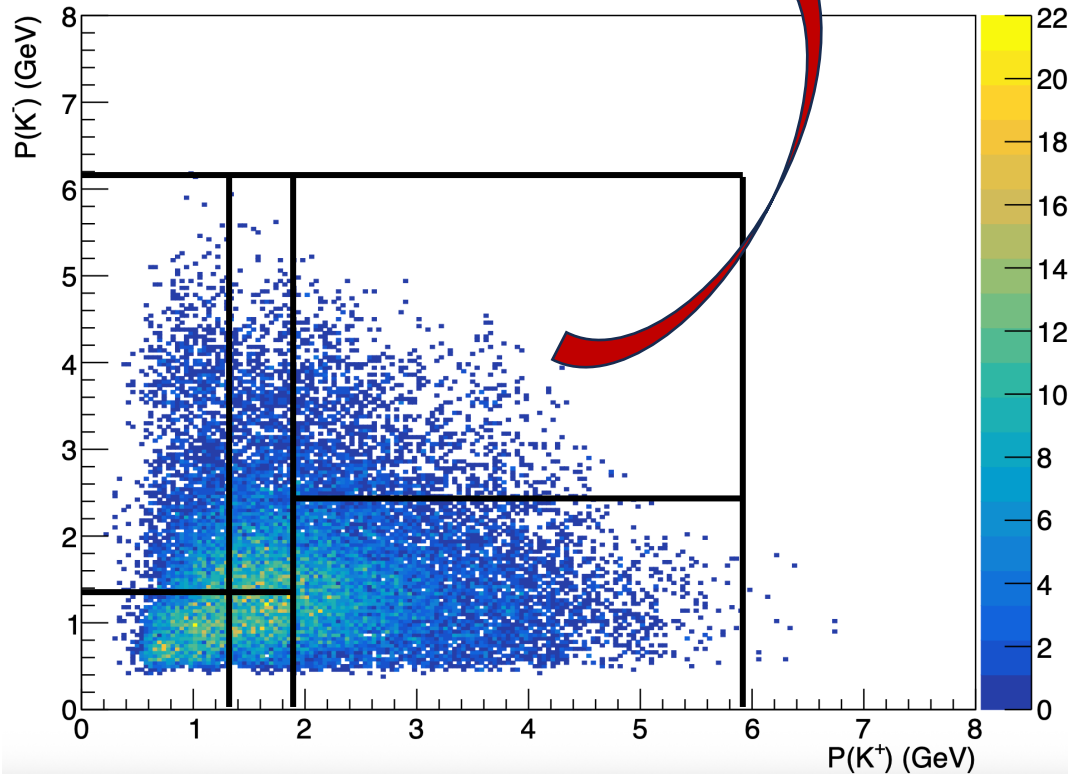
MC

Random fill from the MC
reconstructed event histogram
(with the smearing which depends
of the parameters $A_1, B_1, C_1 \dots$)

Random fill from the
distribution of data
background

Smearing of the kaons

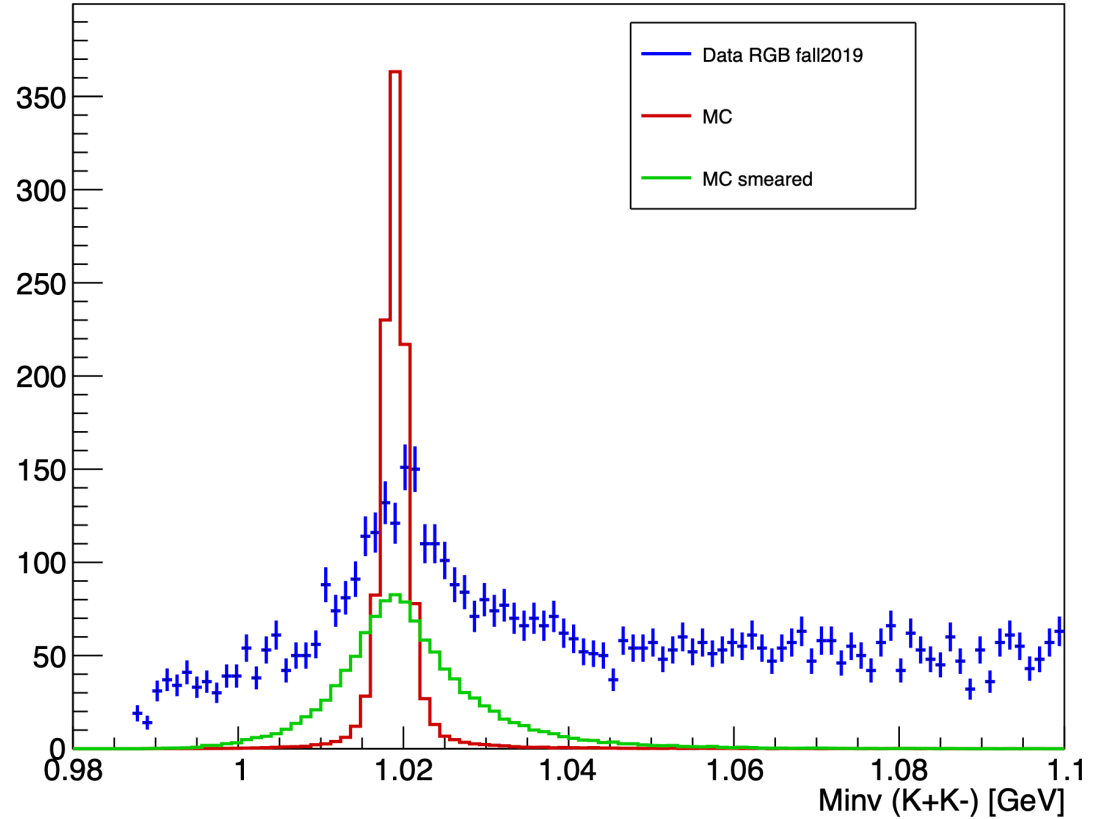
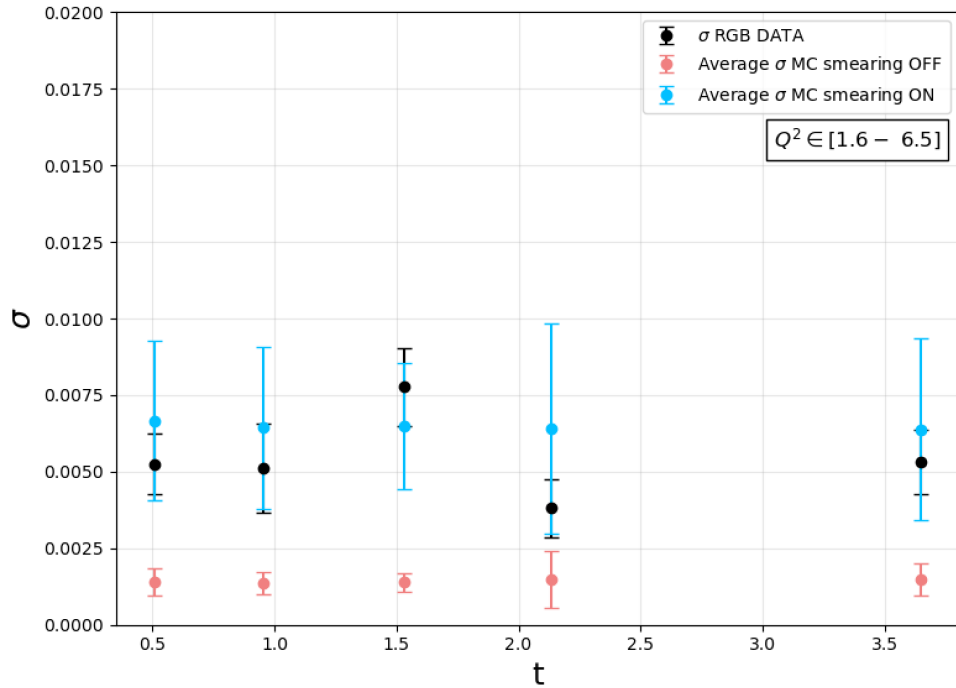
Extract σ_{DATA} and σ_{MC} per bins



$$X^2 = \sum_{bins} \sigma_{DATA} - \sigma_{MC}$$

→ Minimise X^2 with Minuit2 (Simplex) and find 8 parameters A1, B1, C1, D1, A2, B2, C2, and D2 of the smearing functions

Smearing of the kaons



Update on the acceptance : new method

$$A = \frac{N_{rec}}{N_{gen}} * \frac{N_{fit}}{N_{random}}$$



Contribution from the model, the geometry and the efficiency.



Contribution from the resolution of the fit.

N_{rec} : Number of reconstructed event in the MC

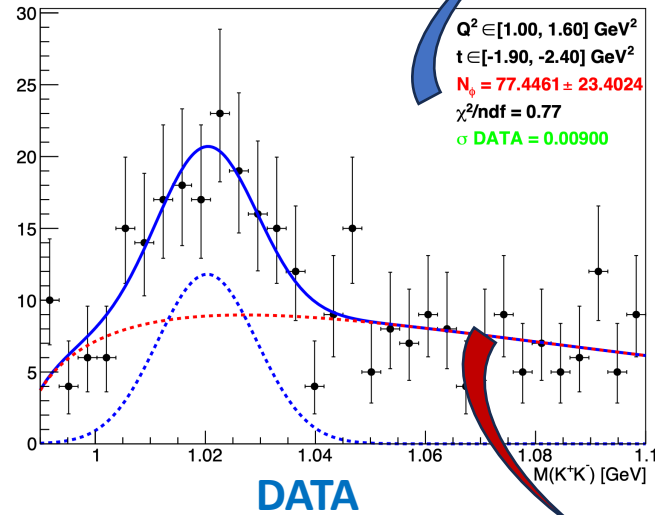
N_{gen} : Number of generated event in the MC

N_{random} : Number of event random fill from the MC reconstructed histogram (equal the number of phi extracted in the data)

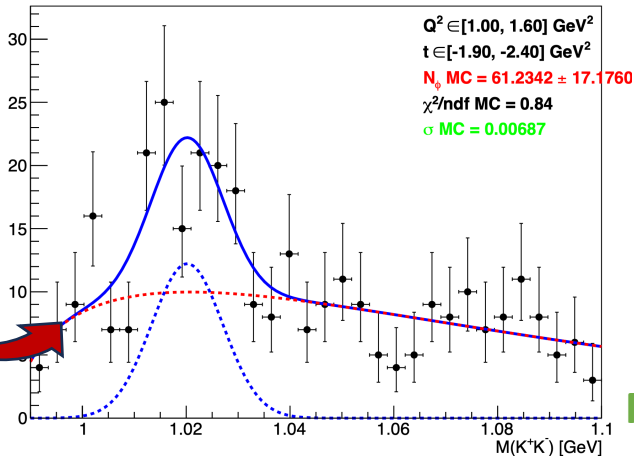
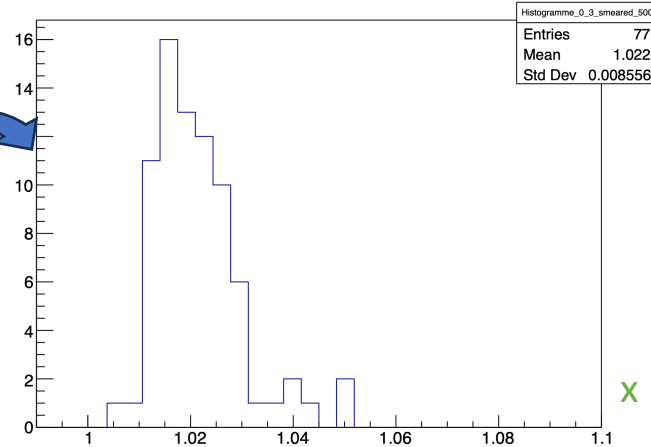
N_{fit} : Number of extracted event with the fit (average of 10000 pseudo-exp)

Update on the acceptance : new method

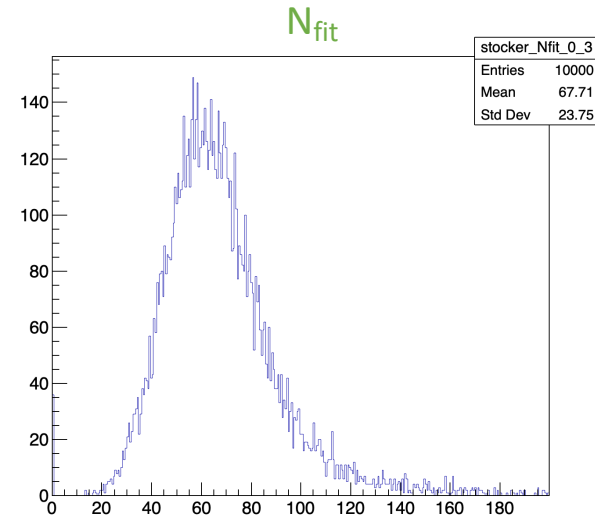
Random fill (with N_{random} equal to the number of ϕ in the data) from the MC reconstructed histogram.



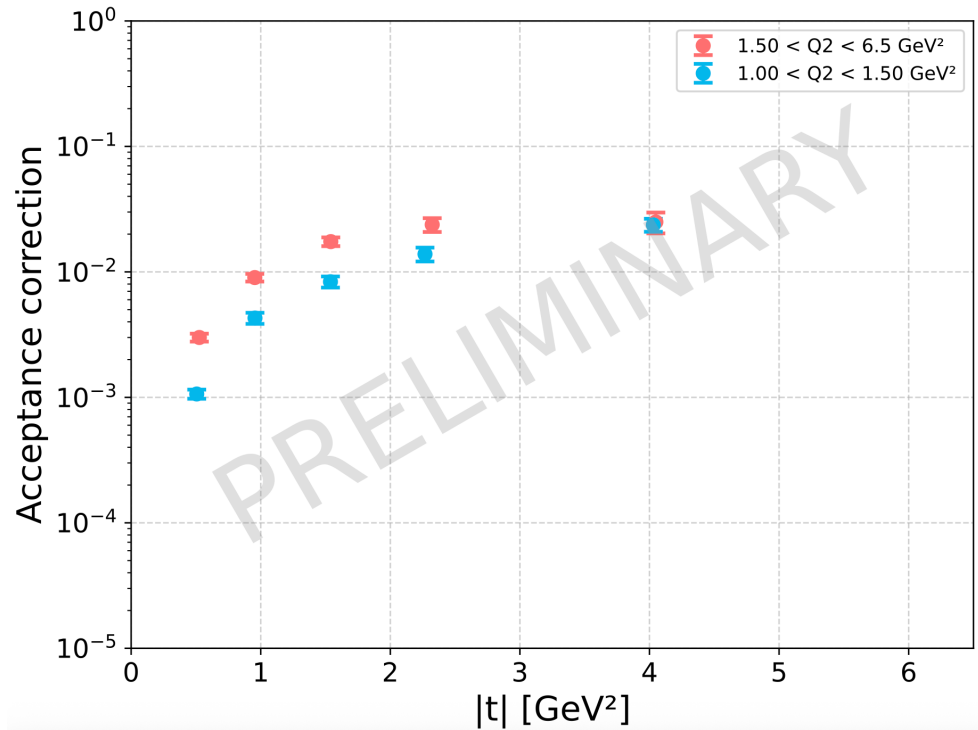
Random fill from the distribution of data background



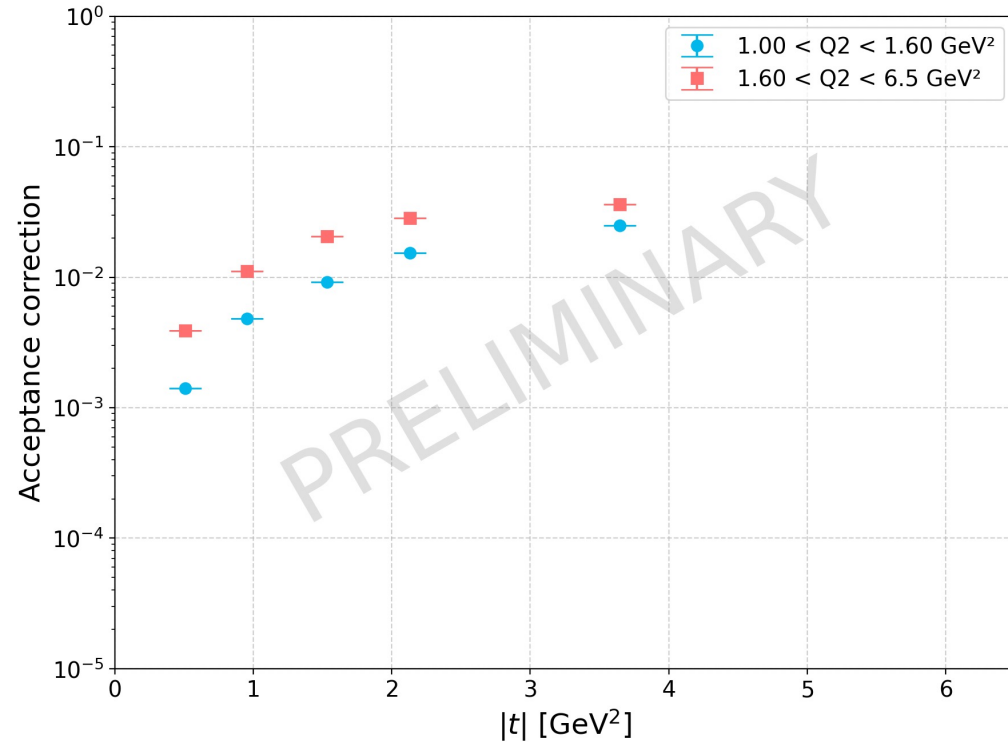
$\times 10\,000$



Update on the acceptance : new method

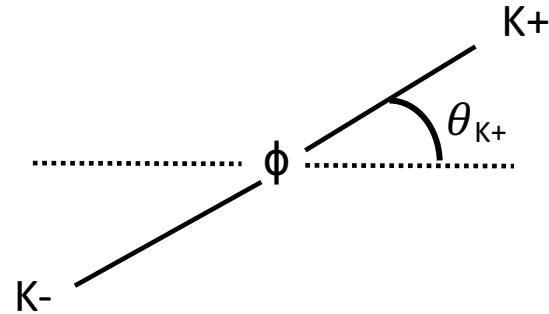
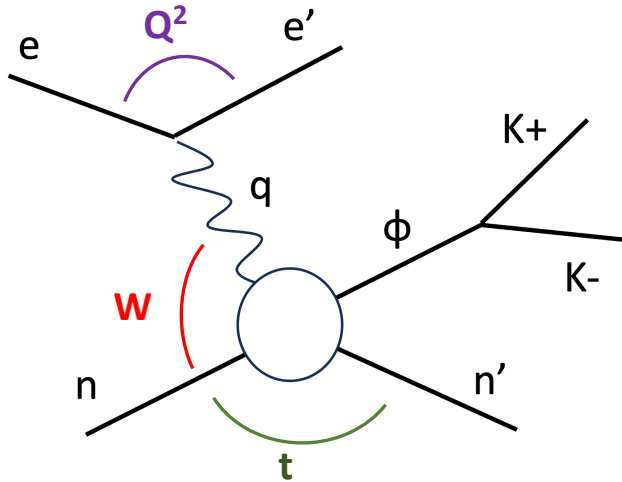


With the **previous** method : $A = \frac{N_{rec}}{N_{gen}}$



With **new** method : $A = \frac{N_{rec}}{N_{gen}} * \frac{N_{fit}}{N_{random}}$
(and smearing)

Include SDME in the simulation

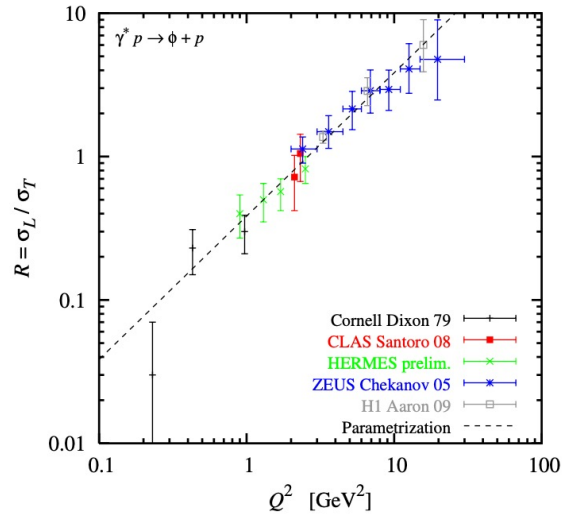


Non isotropic decay of the kaons in the ϕ frame. So we need to add a weight associated to the SDME

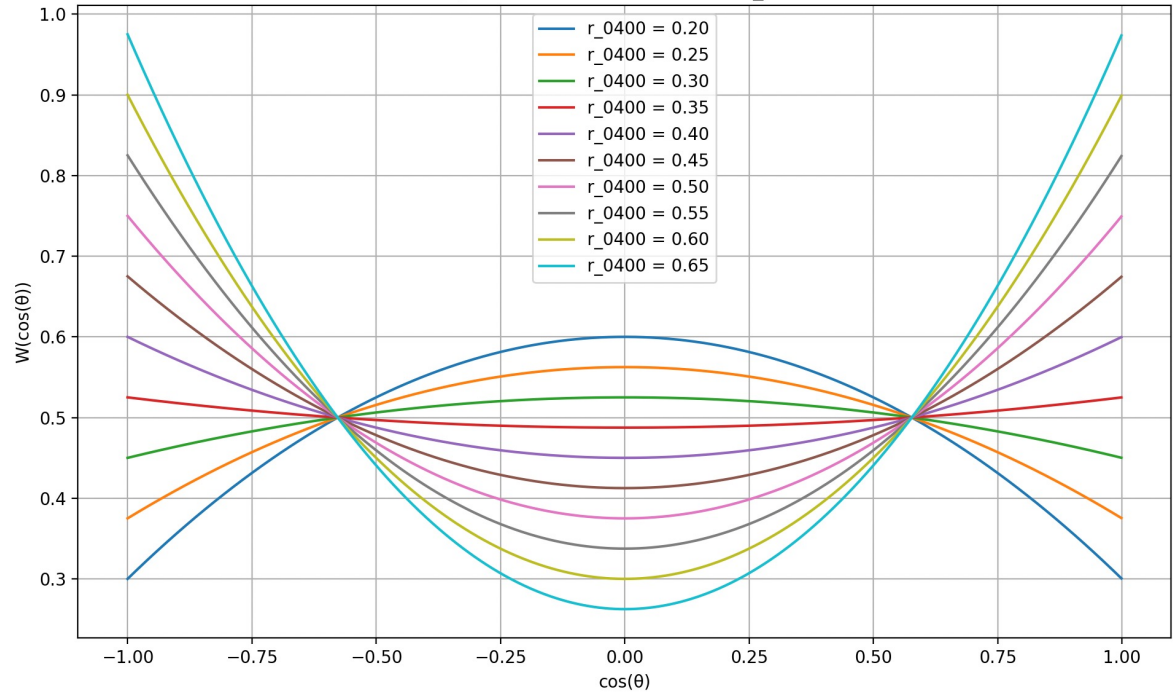
$$Total\ weight = Phase\ Space * \frac{d^3\sigma}{dt dQ^2 dx_b} * SDME * BR_{\phi \rightarrow K^+ K^-}$$

Include SDME in the simulation

$$W(\cos \theta_H) = \frac{3}{4} \left[\left(1 - r_{00}^{04}\right) + \left(3r_{00}^{04} - 1\right) \cos^2 \theta_H \right]$$

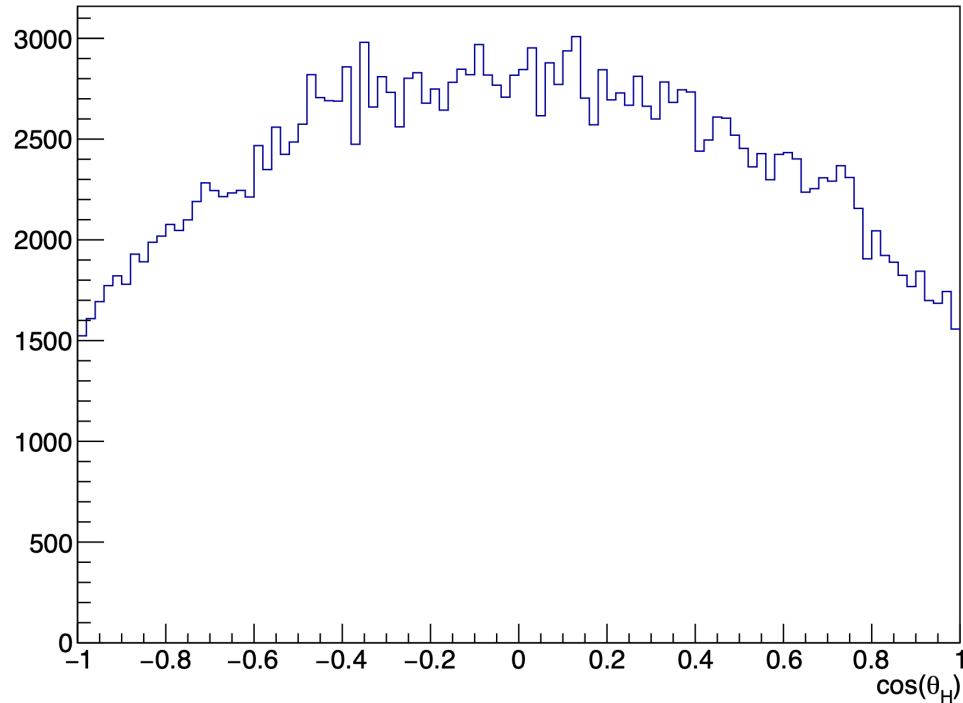


$$R = \frac{r_{00}^{U4}}{\epsilon(1 - r_{00}^{04})} \quad R(W, Q^2) = \frac{c_R Q^2}{m_\phi^2}$$

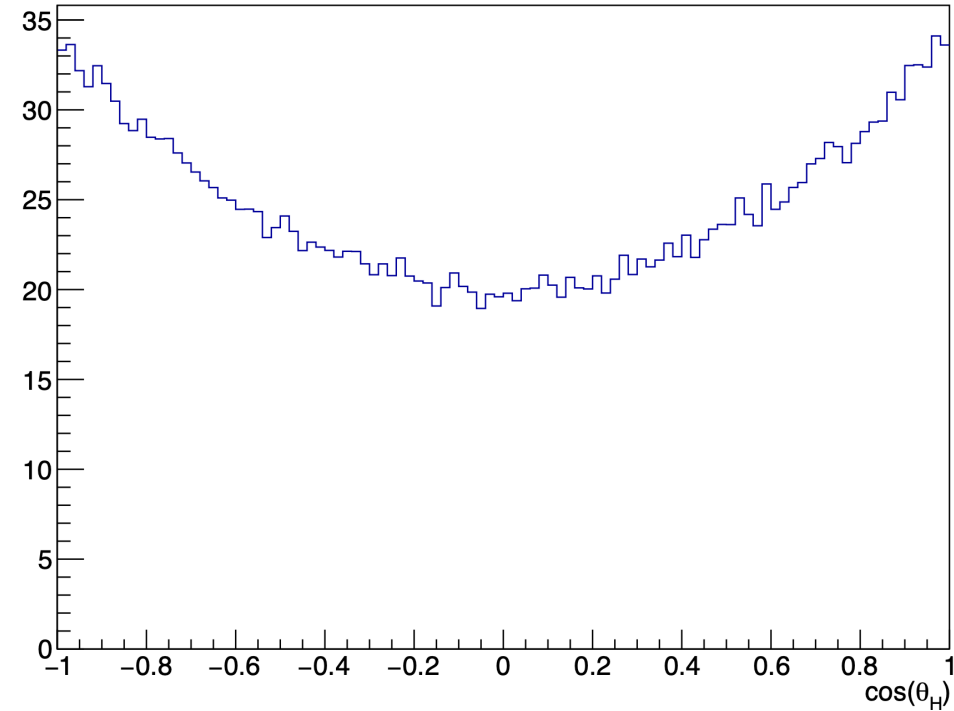


From Proposal to Jefferson Lab PAC39 Exclusive Phi Meson
Electroproduction with CLAS12

Include SDME in the simulation (generated plots)



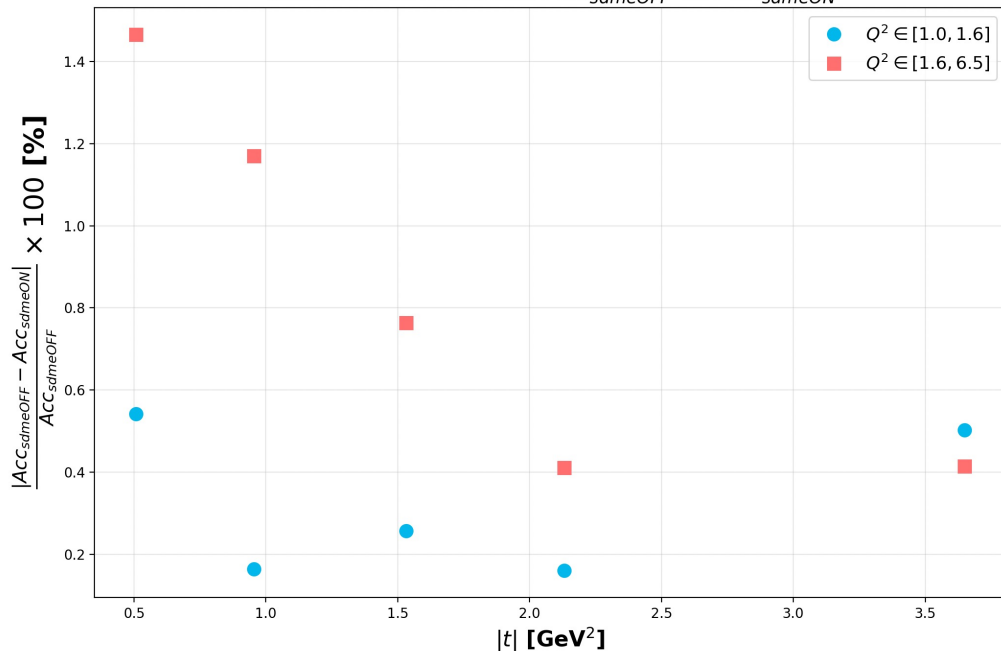
With Q^2 in $[1.0 - 1.5] \text{ GeV}^2$



With Q^2 in $[5.0 - 6.5] \text{ GeV}^2$

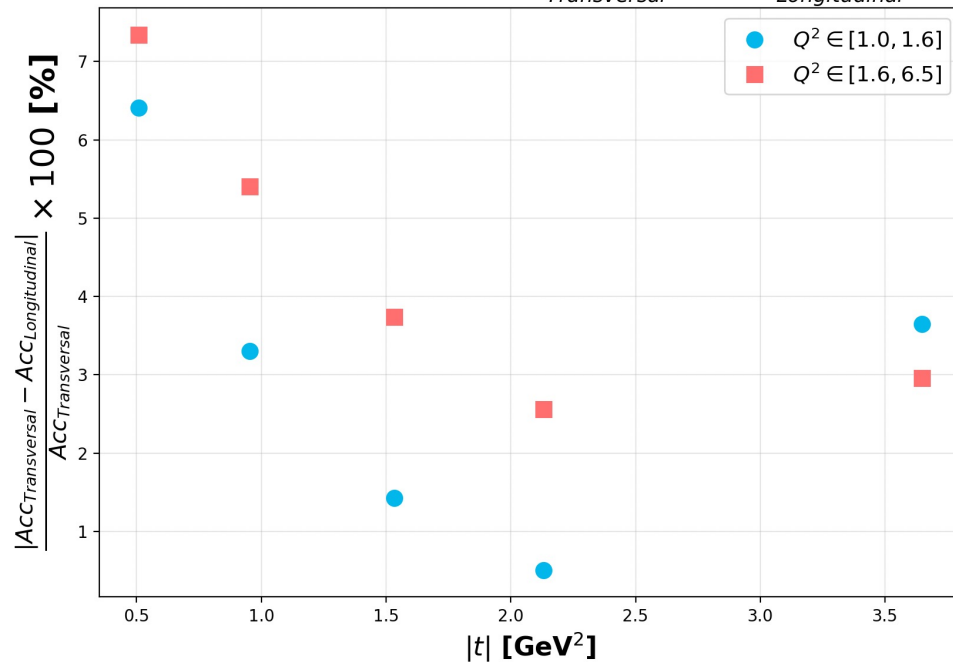
Impact on the acceptance

Relative difference: $ACC_{sdmeOFF}$ vs ACC_{sdmeON}



Impact < 1.5 % on the acceptance between no SDME and a classical model of SDME

Relative difference: $ACC_{Transversal}$ vs $ACC_{Longitudinal}$



Impact < 7.5 % between two different models of SDME (longitudinal or transversal favoured distributions)

Preliminary differential cross section with all update

The differential cross section of this reaction $e n \rightarrow e' n' \phi$ is :

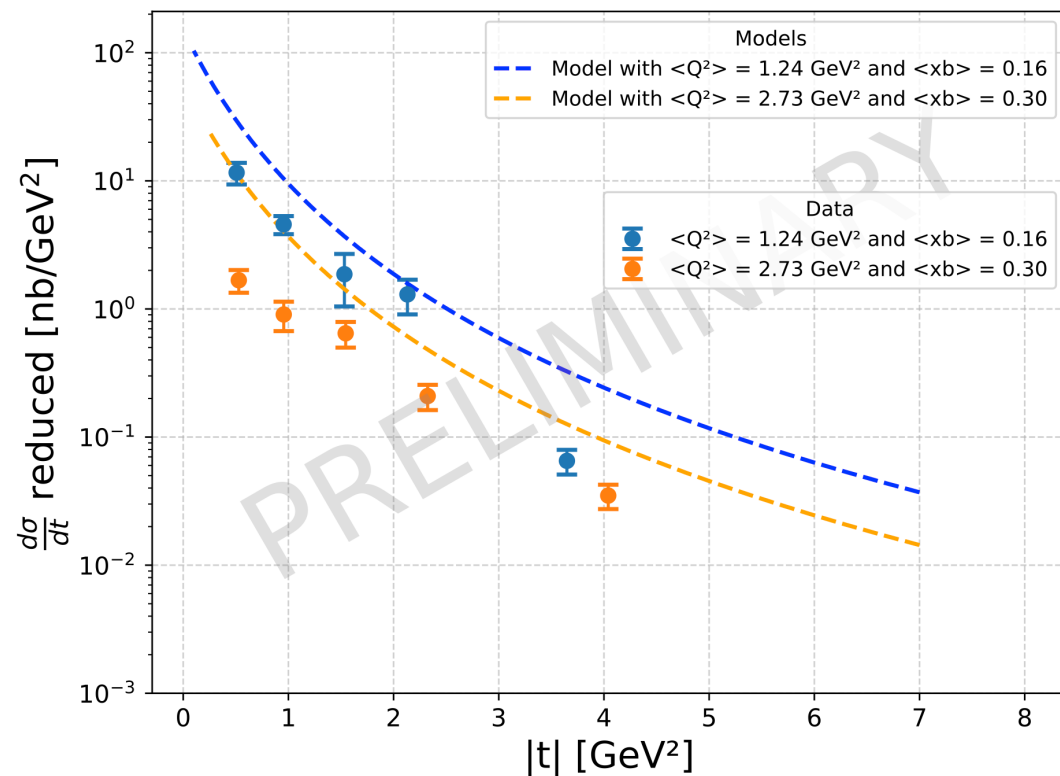
$$\frac{d^3\sigma}{dQ^2 dx_B dt} = \Gamma(Q^2, x_B, E) \left[\frac{d\sigma_T}{dt}(Q^2, x_B, t) + \epsilon \frac{d\sigma_L}{dt}(Q^2, x_B, t) \right]$$

with $\Gamma \equiv \frac{\alpha}{8\pi} \frac{Q^2}{m_N^2 E^2} \frac{1-x_B}{x_B^3} \frac{1}{1-\epsilon}$ the virtual photon flux factor.

The reduced differential cross section of this reaction $\gamma^* n \rightarrow n' \phi$ is given by :

$$\frac{d\sigma}{dt}(Q^2, x_B) = \frac{1}{\Gamma(Q^2, x_B, E)} \cdot \frac{d^3\sigma}{dQ^2 dx_B dt}$$

$$\frac{d\sigma}{dt}(Q^2, x_B) = \frac{\Delta N}{\Gamma \cdot \Delta x_B \cdot \Delta Q^2 \cdot \Delta t \cdot f \cdot L \cdot Acc \cdot Br}$$



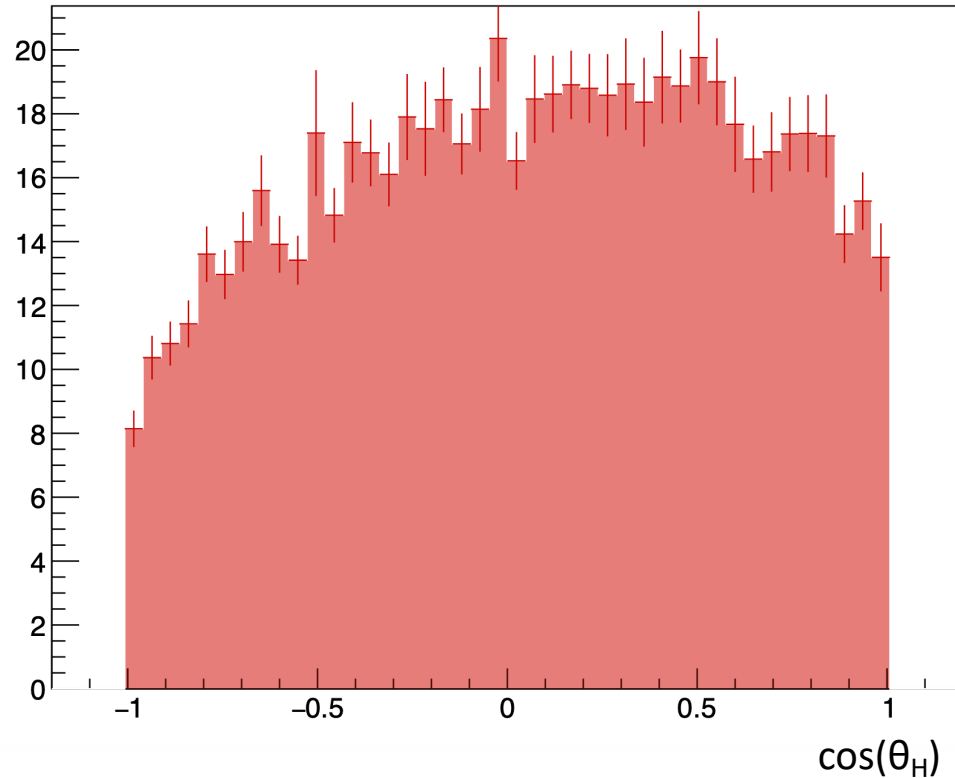
Next steps in the ϕ analysis :

- Improve the MC simulations (fermi motion, efficiency corrections for the kaons...)
- Develop momentum corrections for the kaons.

Thanks!

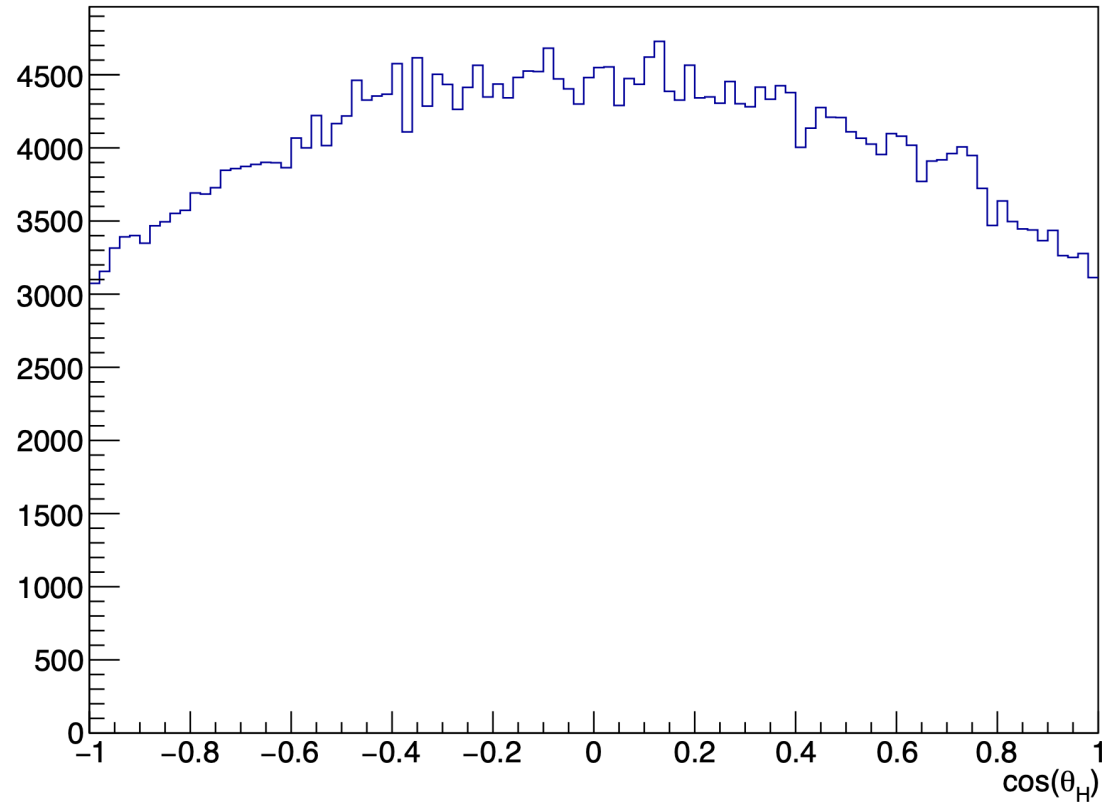
Backup slides

Include SDME in the simulation (reconstructed event)



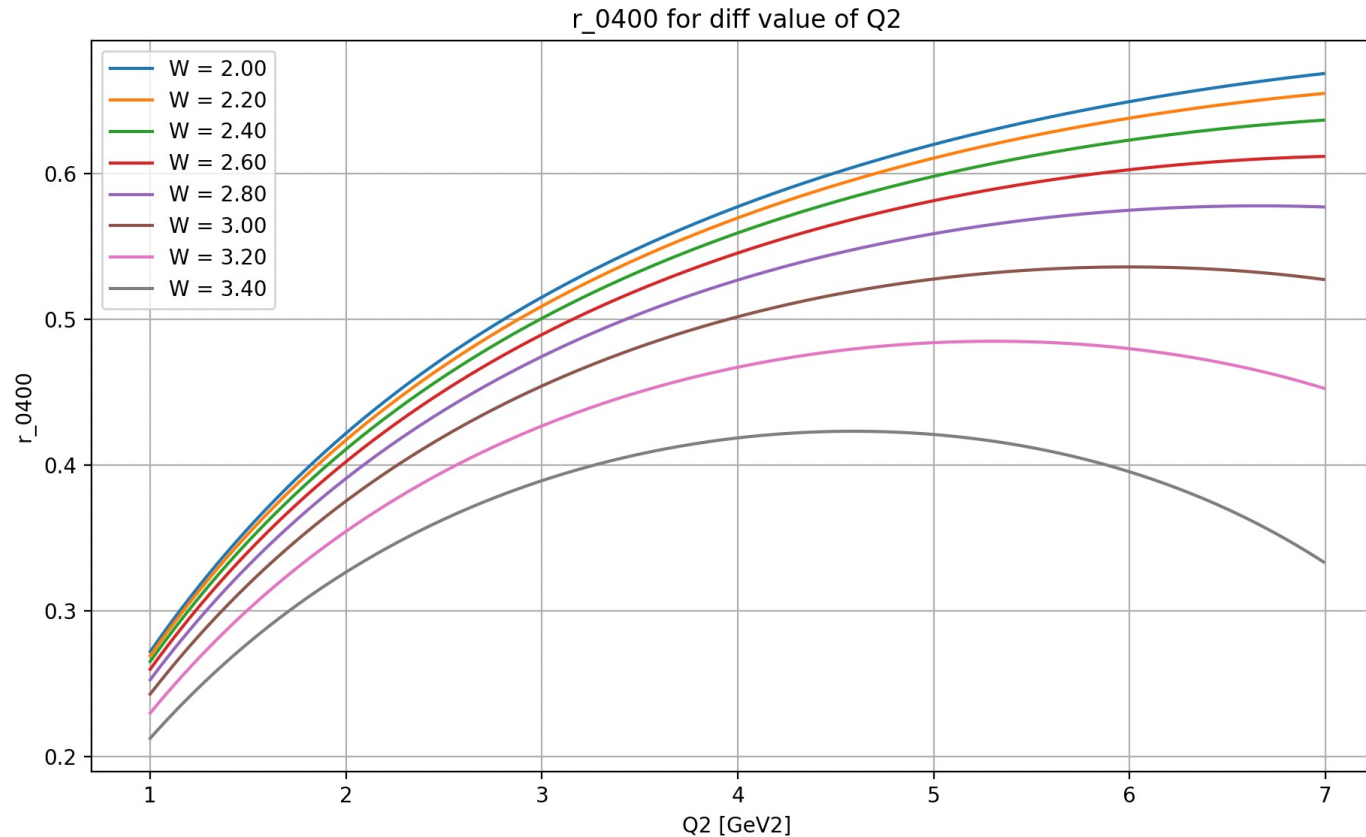
- Reconstructed with Q^2 in $[1.0 - 1.5] \text{ GeV}^2$
- You can see acceptance effect when you compare to the generated plot

Include SDME in the simulation (generated plots)



With no cut on Q^2

Include SDME in the simulation (generated plots)

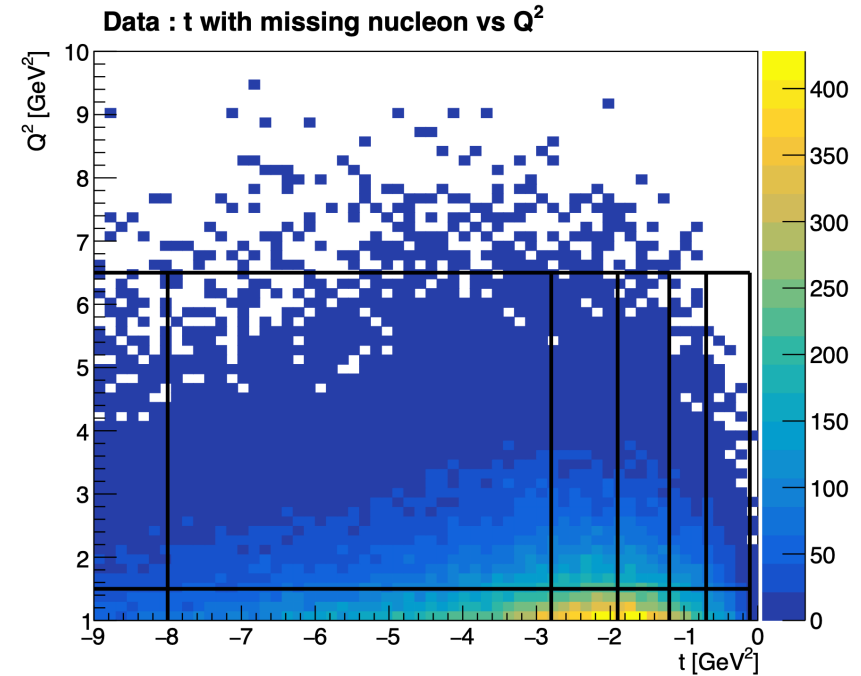
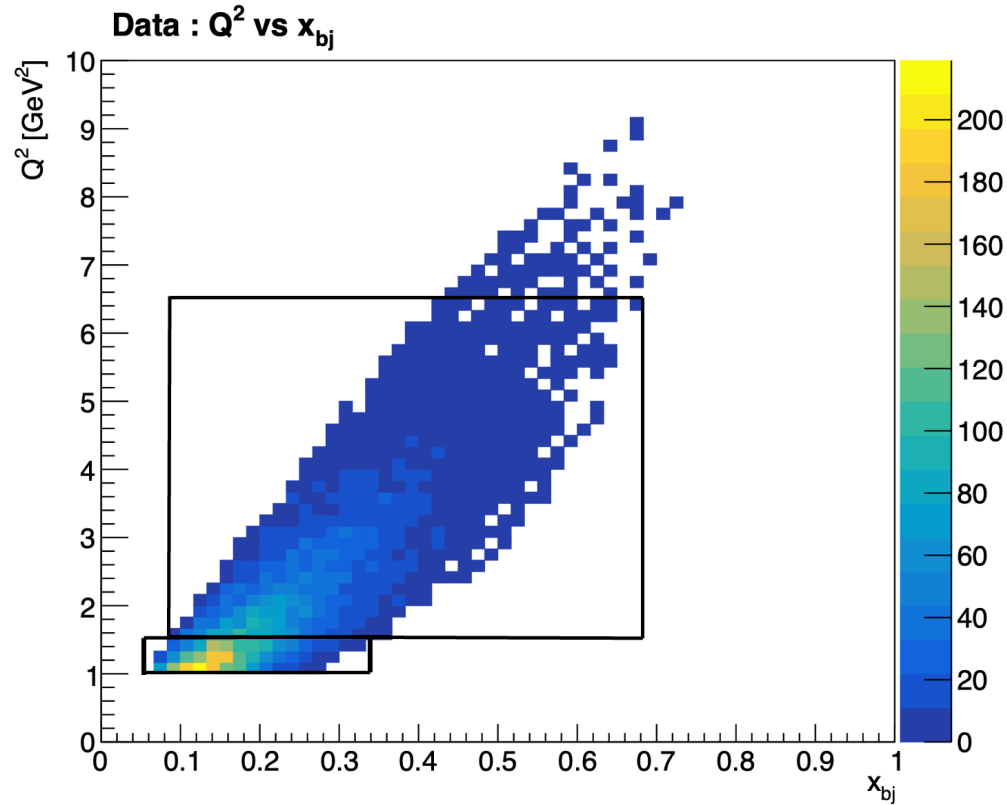


Event selection : second step



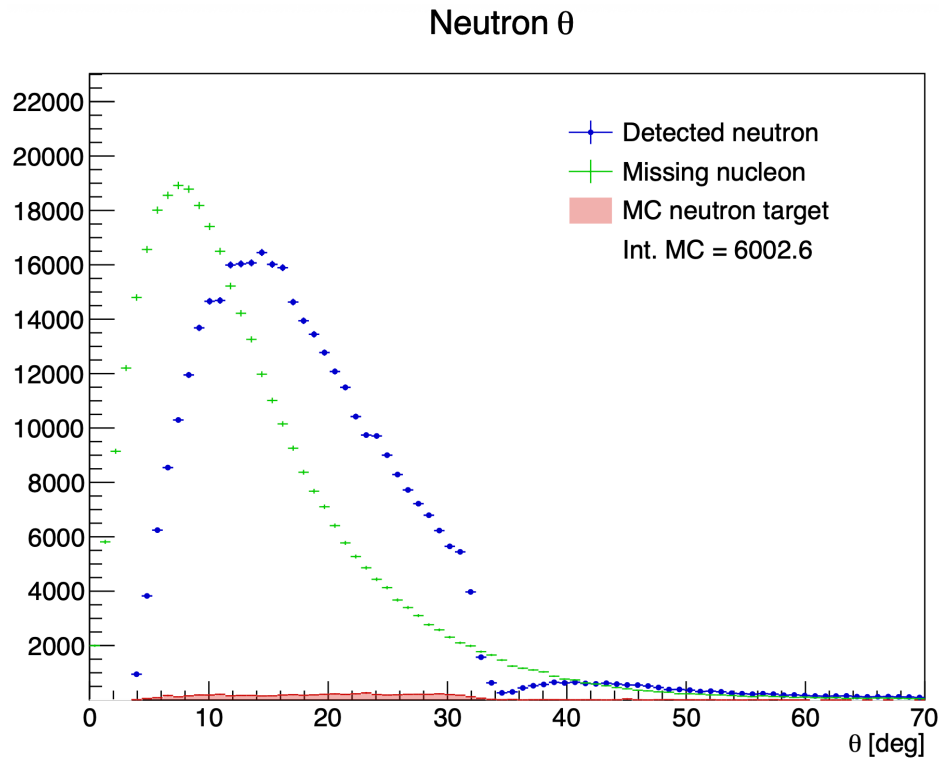
- Apply a cut on the angle between **the missing nucleon** and the **neutron** $< 5^\circ$
- The **missing nucleon** is use to calculate $t = (\text{target} - \text{missing})^2$
- Also applu a cut on $-0.5 < M_{eK^+K^-nX}^2 < 0.5 \text{ GeV}^2$
- Ask for 0 proton detected in the event to avoid proton contamination

Binning of the data

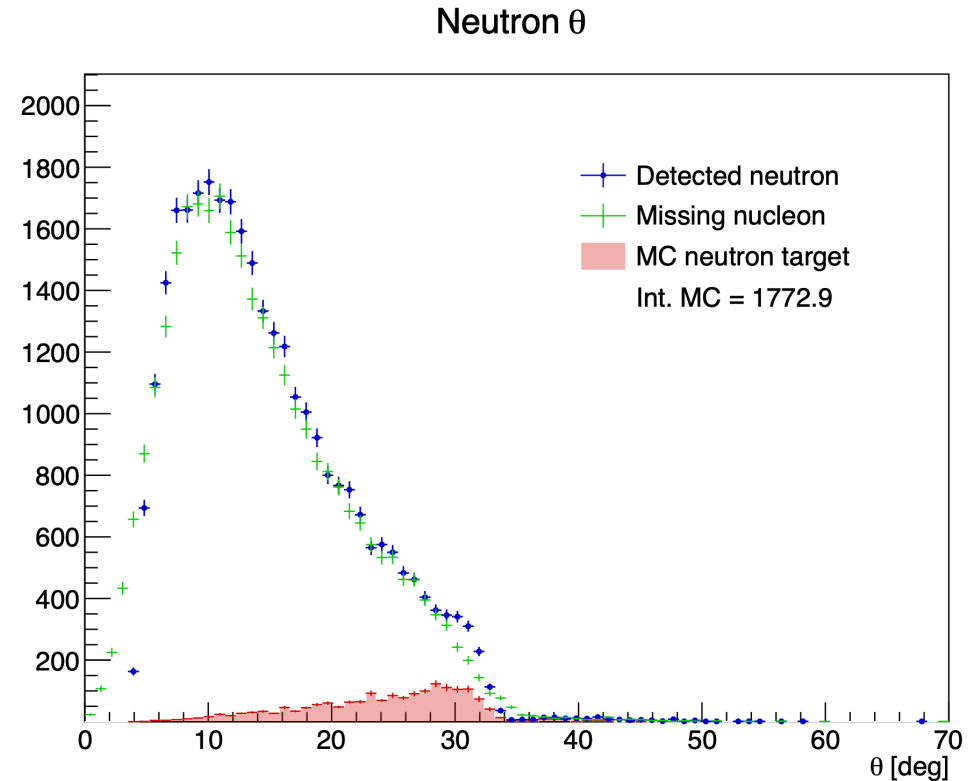


• t binning for the two bins in Q^2

Cut on the angle between missing nucleon and detected neutron



Without the angle cut

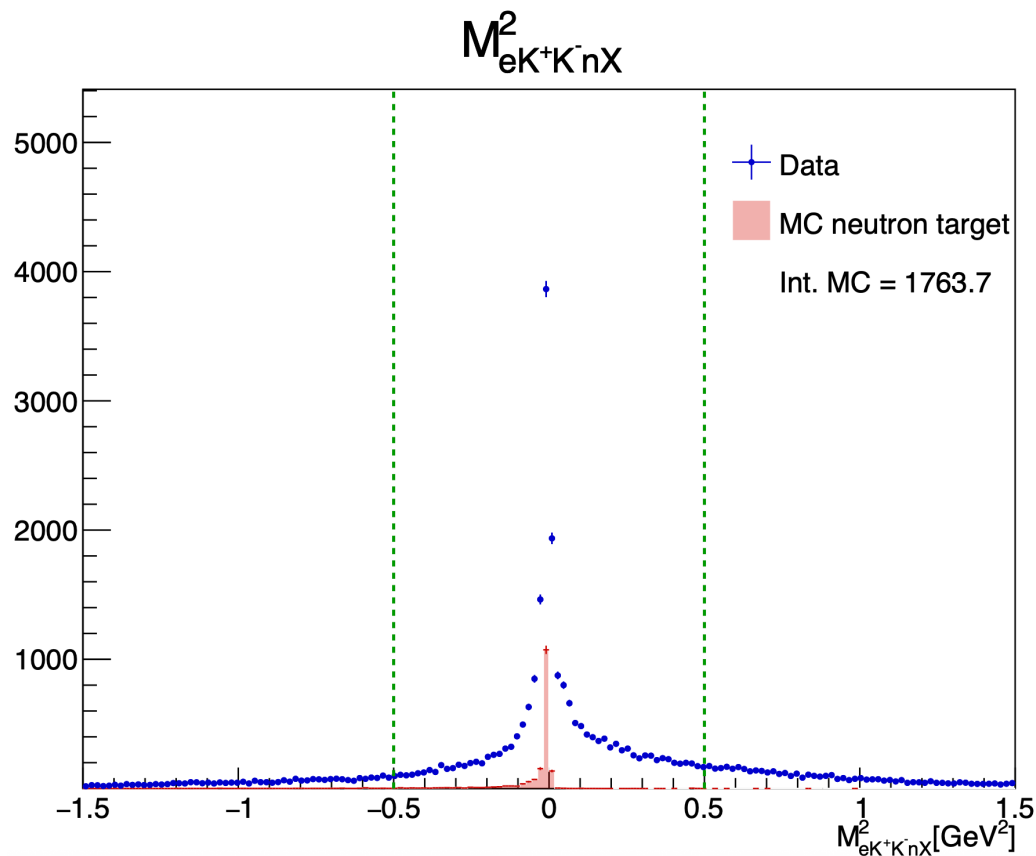


With the angle cut

Cut on total missing mass

Total missing mass $e n \rightarrow e' n' K^+ K^- X$
(after all the previous cut)

$$-0.5 < M_{eK^+K^-nX}^2 < 0.5 \text{ GeV}^2$$



Yield extraction per bins

Fit performed using RooFit.

Signal model :

The signal is described by a Gaussian function :

$$S(M_{inv}) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(M_{inv}-\mu)^2}{2\sigma^2}\right)$$

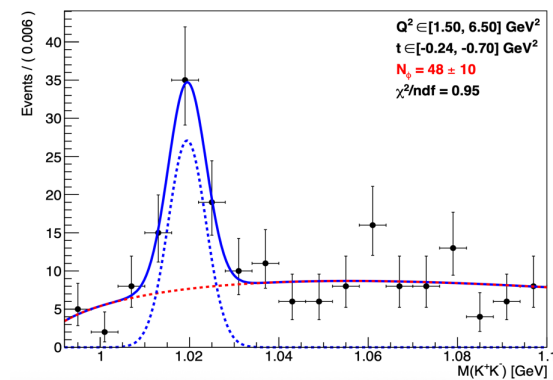
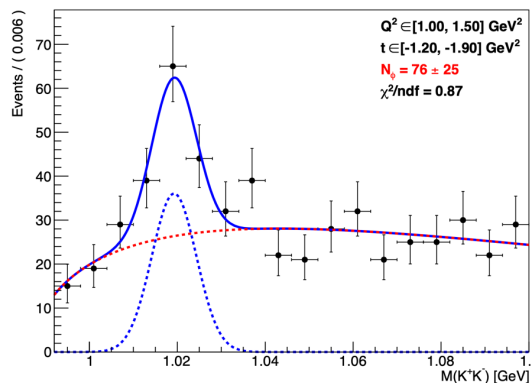
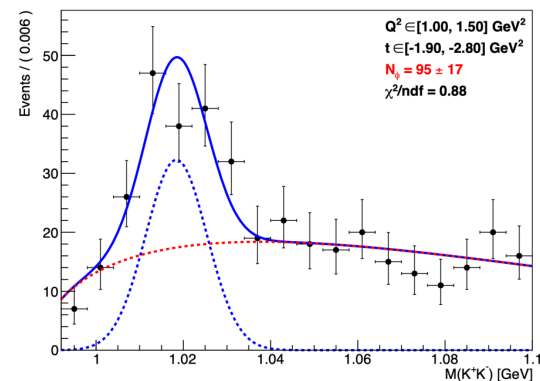
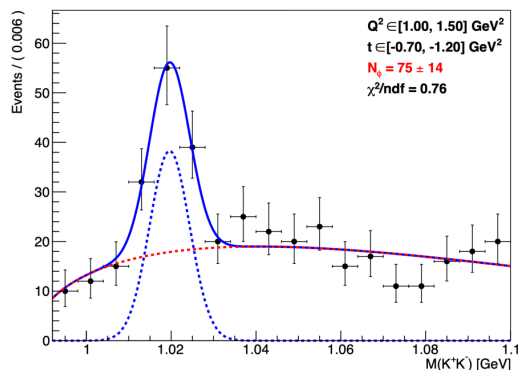
Background model :

The background is modeled using a threshold function multiplied by an exponential polynomial :

$$B(M_{inv}) = \sqrt{M_{inv}^2 - 4m_K^2} \exp(a_0 m + a_1 m^2)$$

where :

- m_K is the kaon mass,
- a_0 and a_1 are free parameters of the fit.

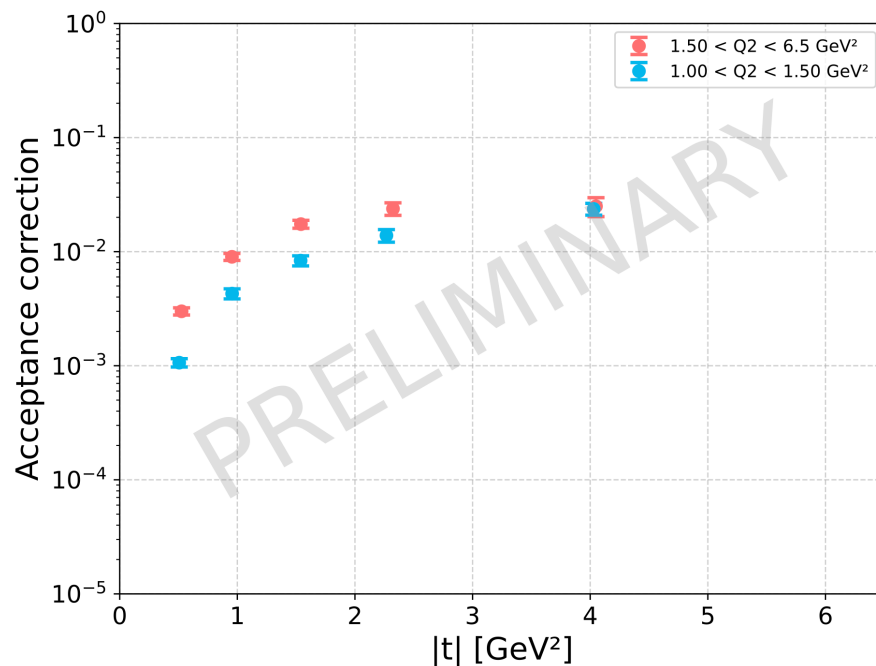
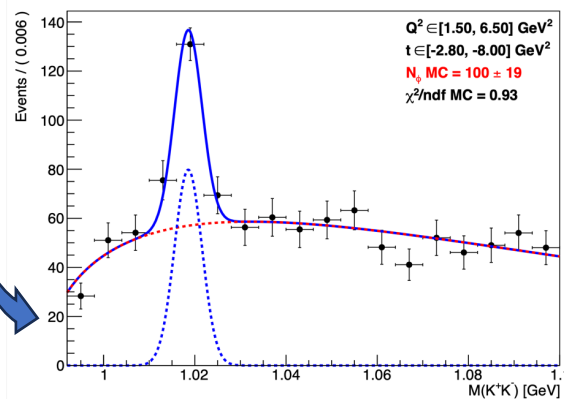
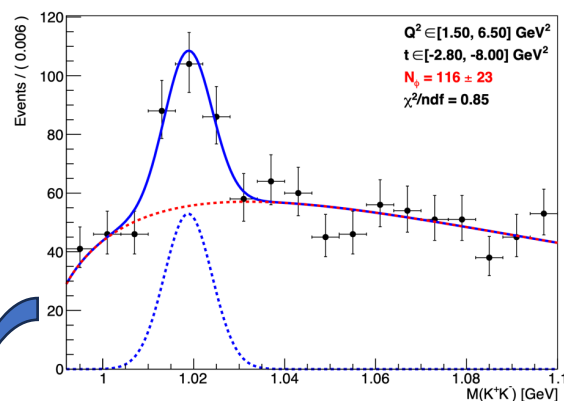


Some examples of fits

Acceptance correction

- Recover the background function from the data and perform a random fill on the MC histogram with a number of events equal to the background integral.

- The objective is to fit the MC and the data under the same conditions.

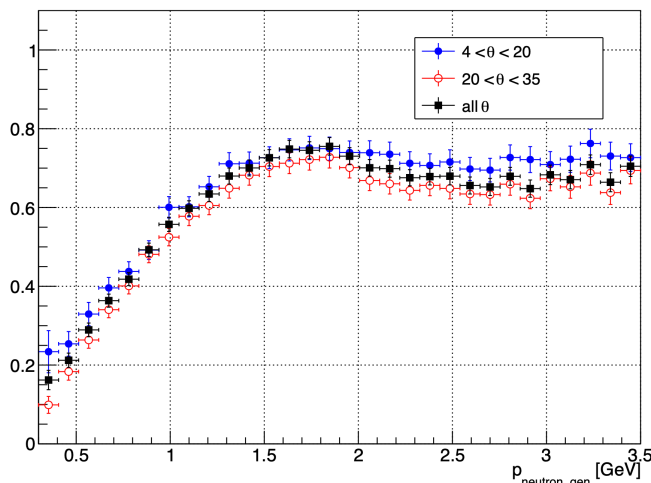


$$A = \frac{N_{rec}}{N_{gen}}$$

- with N_{rec} the number of reconstructed events in MC
- with N_{gen} the number of generated events in MC

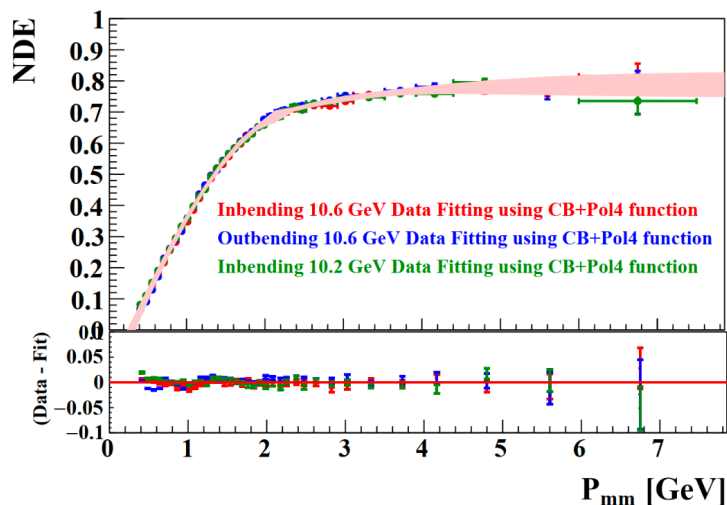
Efficiency correction for neutron in the FD

MC Efficiency vs p



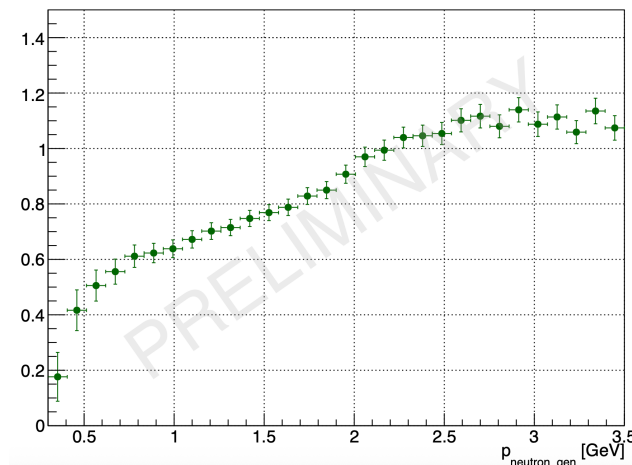
- Neutron efficiency in **MC** : $\frac{N_{rec}(e^- n)}{N_{gen}(\text{with } e^- \text{ rec})}$
- We are in the fiducial volume for reconstructed and genrated event

Neutron Detection Eff



- Neutron efficiency (FD) in **data** from **Lamya Baashen** thesis
(with P_{mm} the missing momentum)

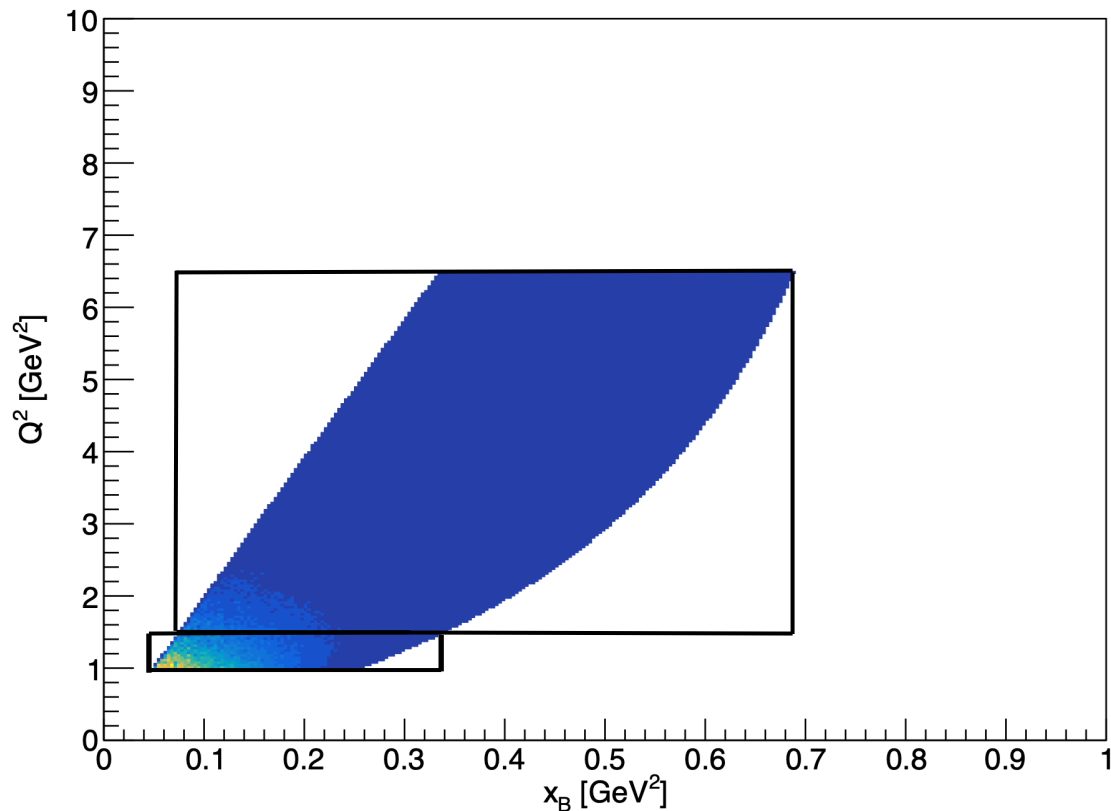
Data/MC Efficiency Ratio



- The ratio between the **MC** and **data** efficiencies gives the **neutron efficiency correction factor**

Binning correction

Generated events : Q^2 vs x_b



- The correction factor f is the size of the physical phase space in Q^2 and x_b divide by $\Delta Q^2 \Delta x_b$
- To determine the size of the physical phase space, we use generated events and divide the number of filled bins by the total number of bins.

bin	$1.0 < Q^2 < 1.5 \text{ GeV}^2$ $0.05 < x_b < 0.34$	$1.5 < Q^2 < 6.5 \text{ GeV}^2$ $0.08 < x_b < 0.7$
f	0.81	0.49

ϕ generator for MC simulation

$$weight_{PhaseSpace} = |Q_{max}^2 - Q_{min}^2| * |xb_{max} - xb_{min}| * |t_{max} - t_{min}|$$

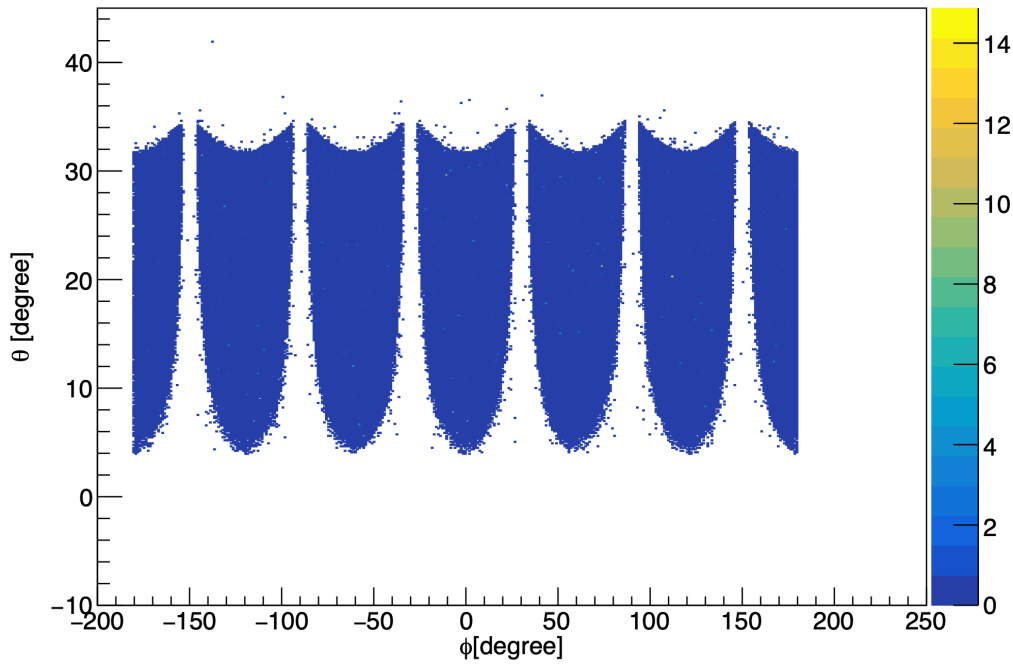
$$\frac{d^3\sigma}{dQ^2 dx_B dt} \quad \text{From Proposal to Jefferson Lab PAC39 Exclusive Phi Meson Electroproduction with CLAS12}$$

$$BR(\phi \rightarrow K^+ K^-) \approx 49.2\%$$

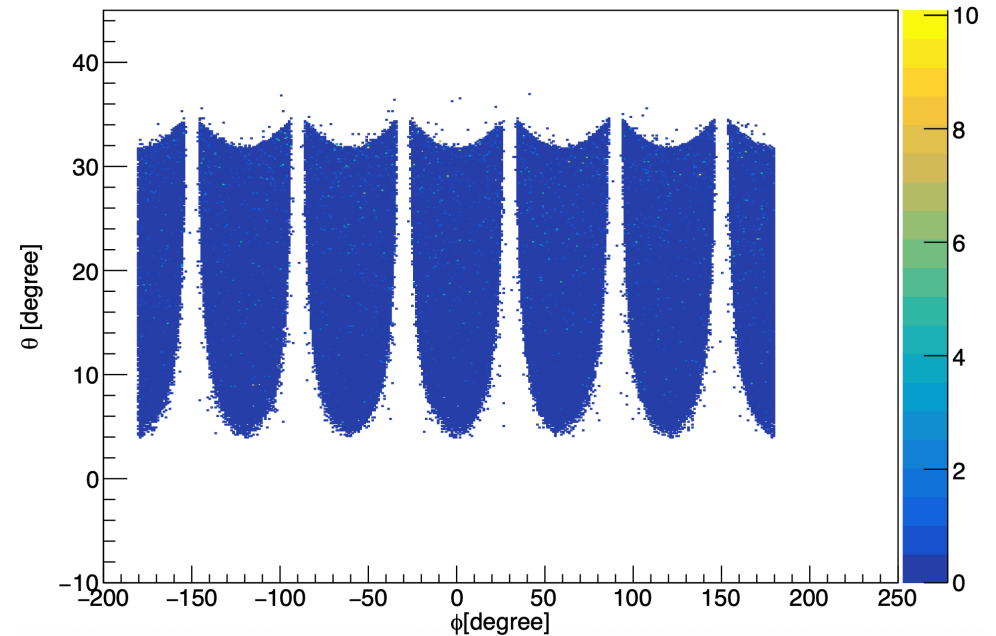
$$totalweight = weight_{phasespace} * weight_{\frac{d^3\sigma}{dt dQ^2 dx_b}} * BR_{\phi \rightarrow K^+ K^-}$$

Neutron efficiency correction additional plots

theta vs phi neutron REC

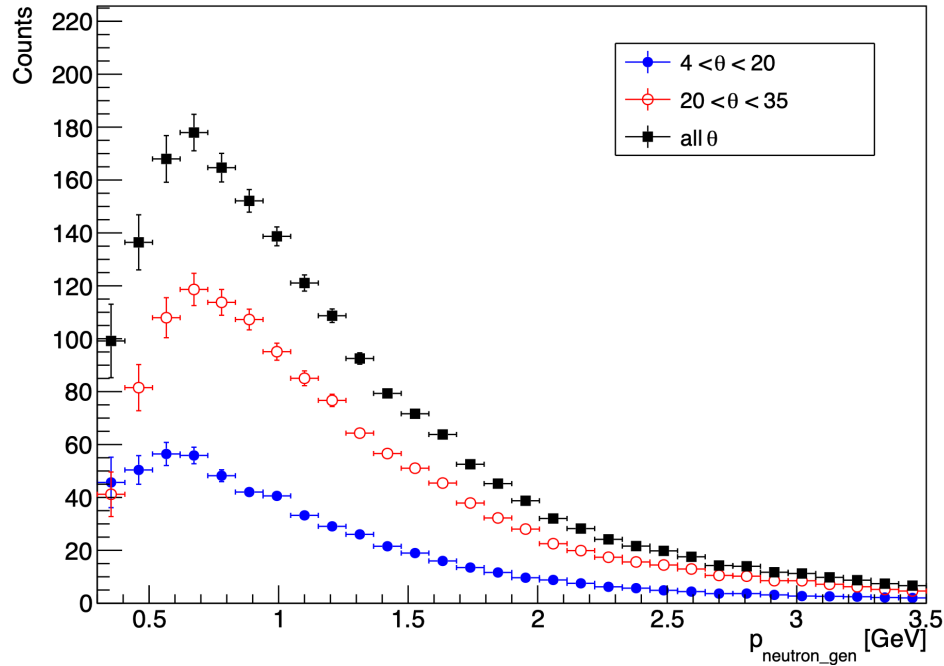


theta vs phi neutron GEN

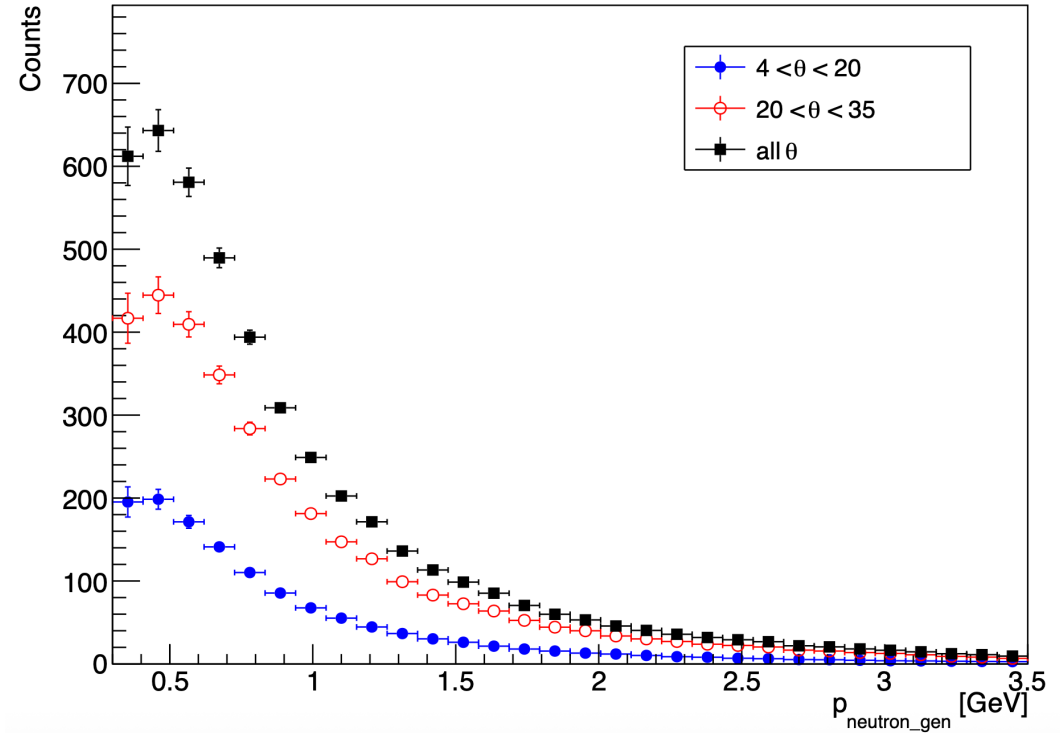


Neutron efficiency correction additional plots

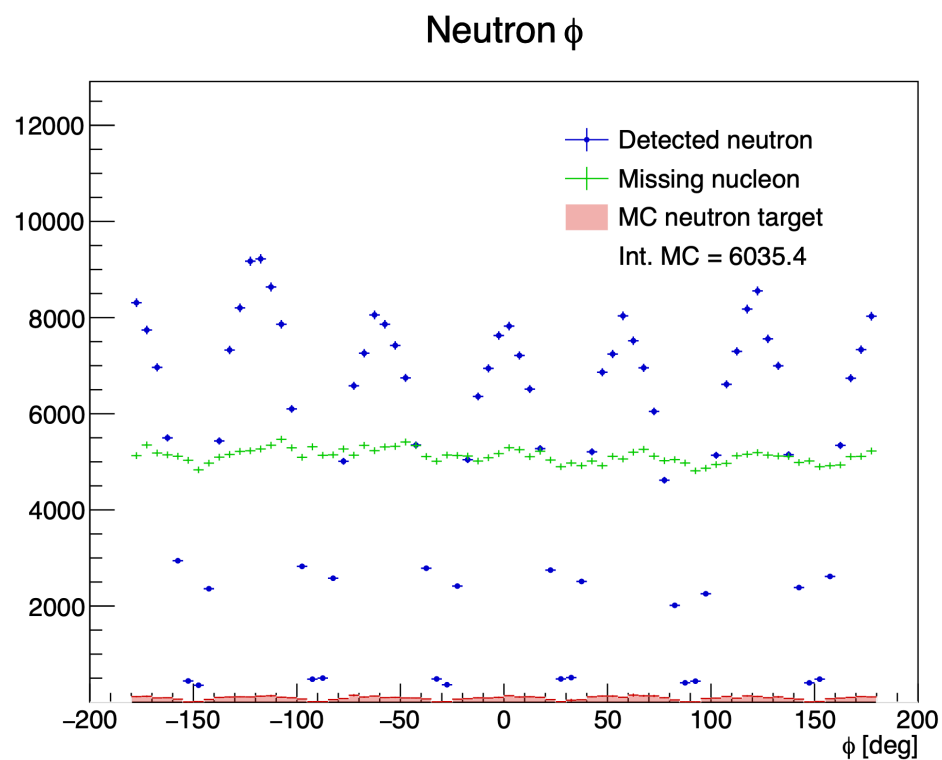
p neutron REC



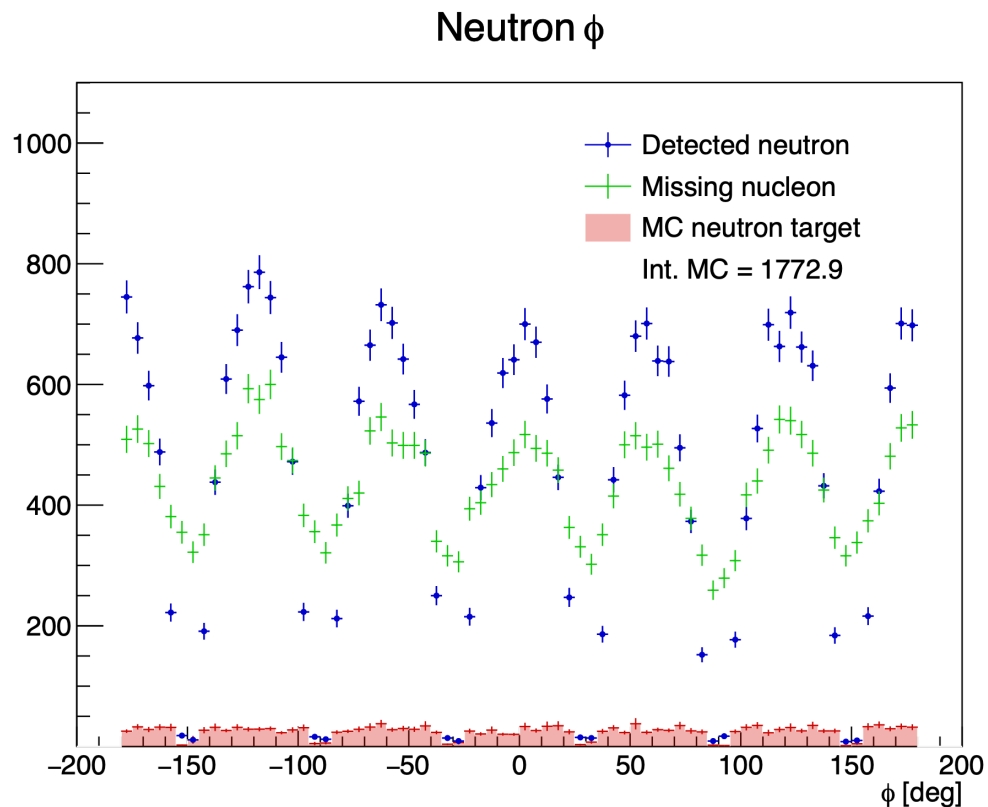
p neutron GEN



Cut on the angle between missing nucleon and detected neutron

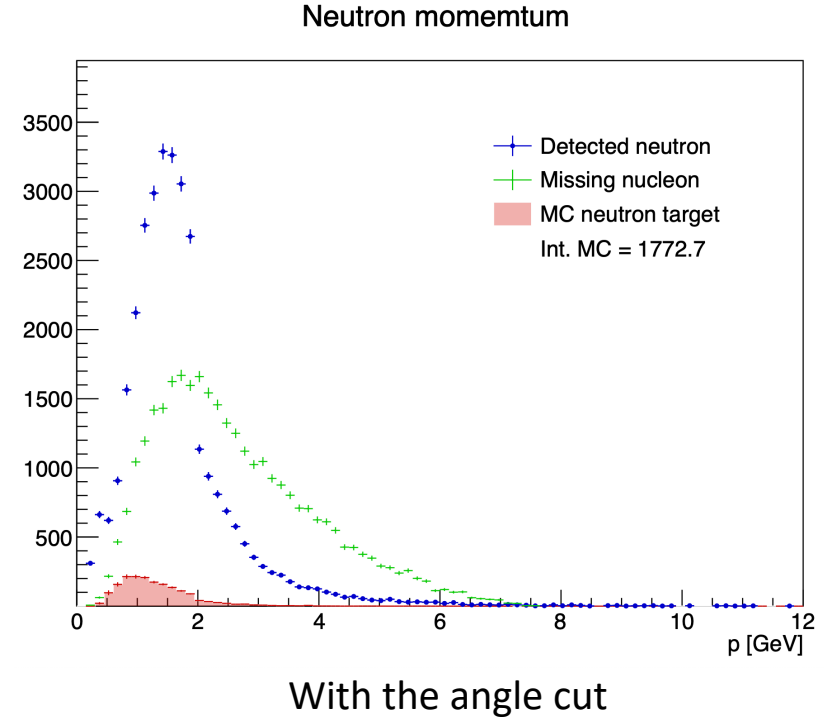
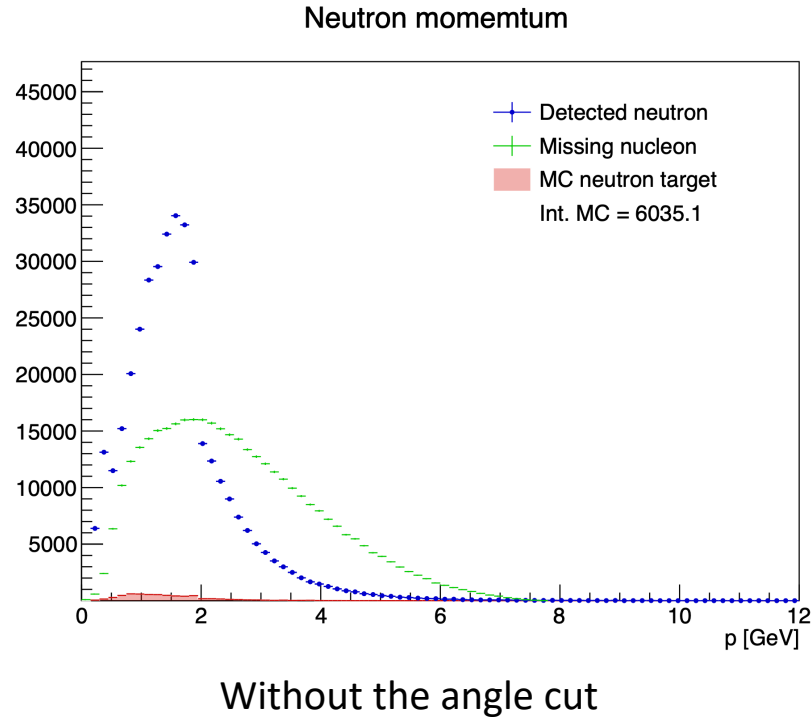


Without the angle cut



With the angle cut

Cut on the angle between missing nucleon and detected neutron



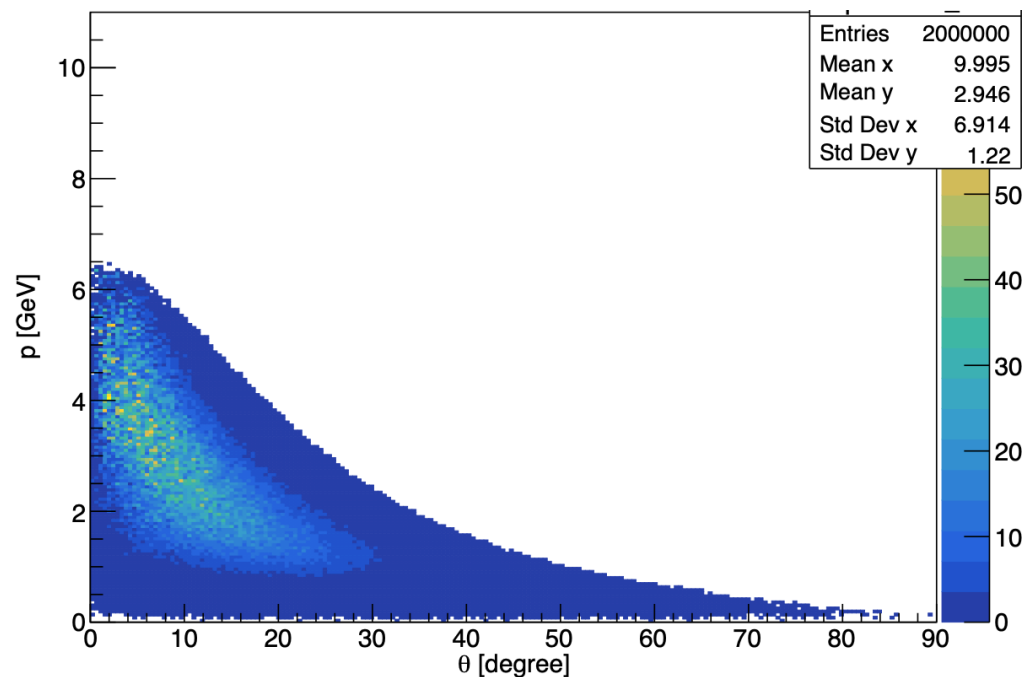
- I do not have a neutron momentum correction yet

Cut on the status of K^+K^-

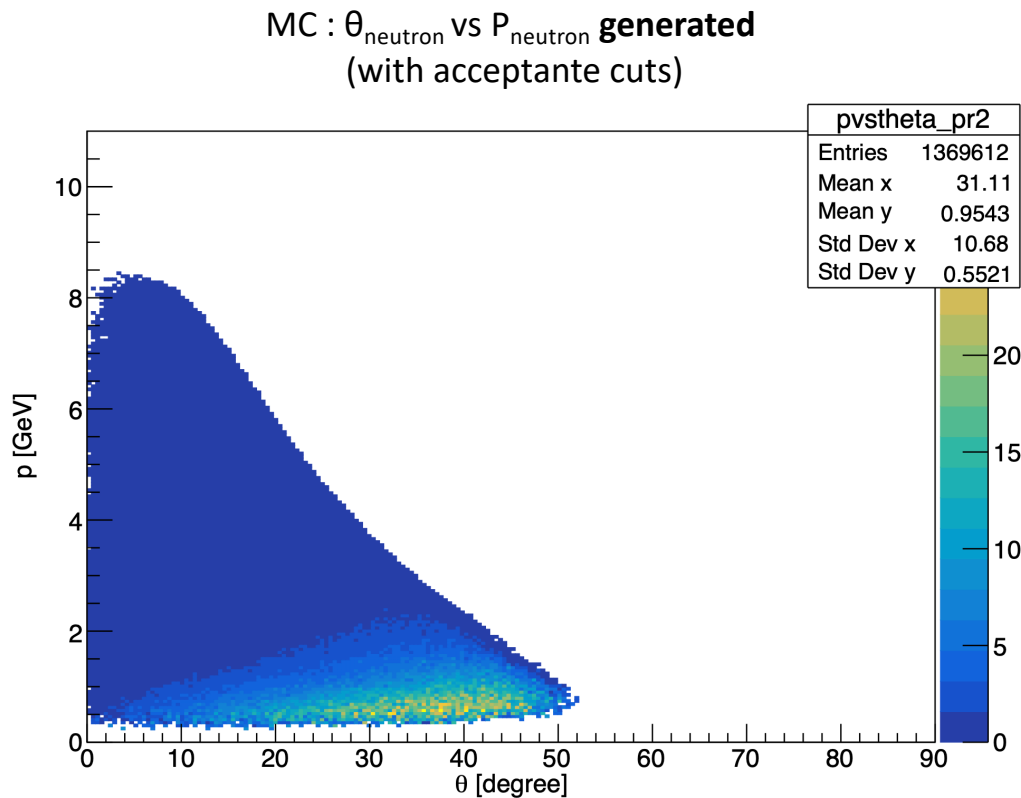
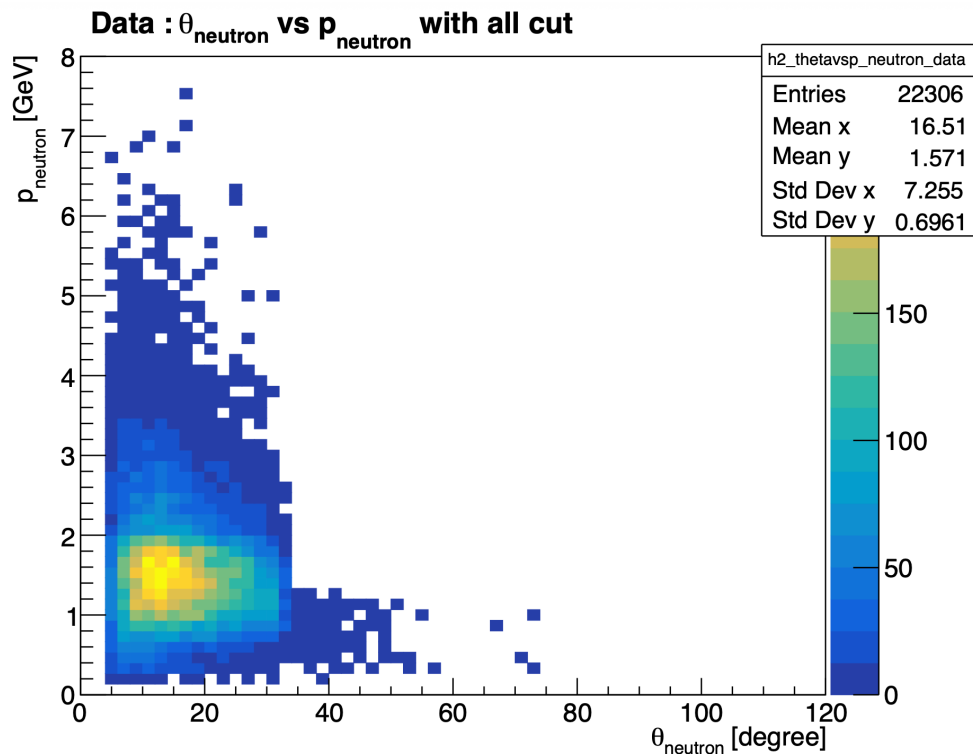
Two justifications to cut $K^+ K^-$ in CD :

- Reconstruction of kaons in the CD is less good than in the FD
- The generator predict few events in the CD

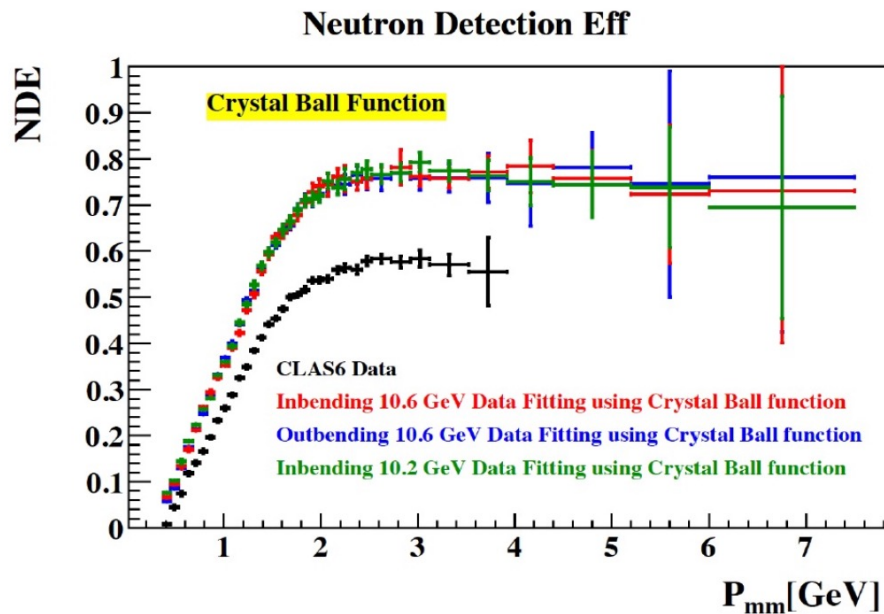
Generated event : p vs theta for K^+



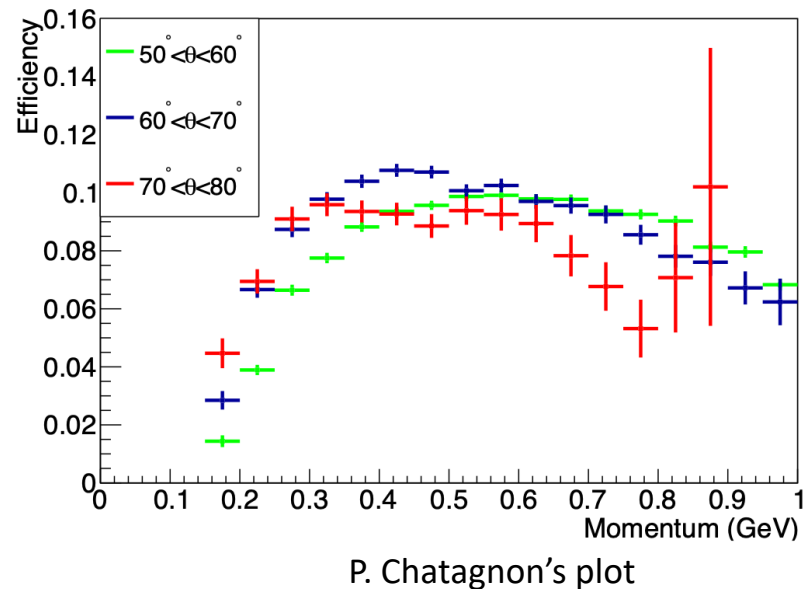
Neutron phasespace :



Neutron phasespace :



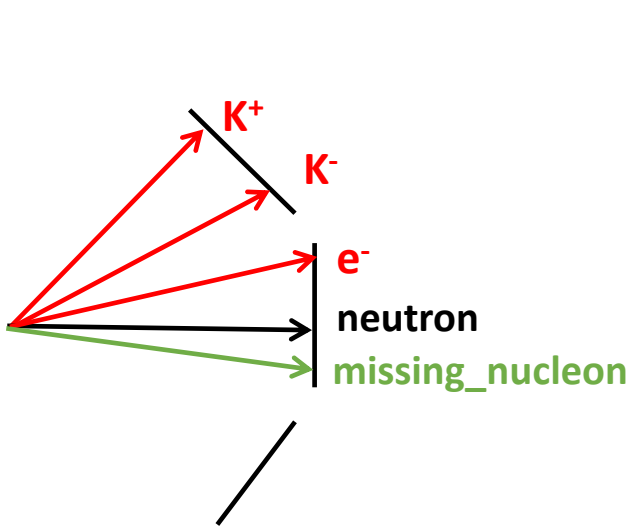
Neutron detection efficiency in the FD



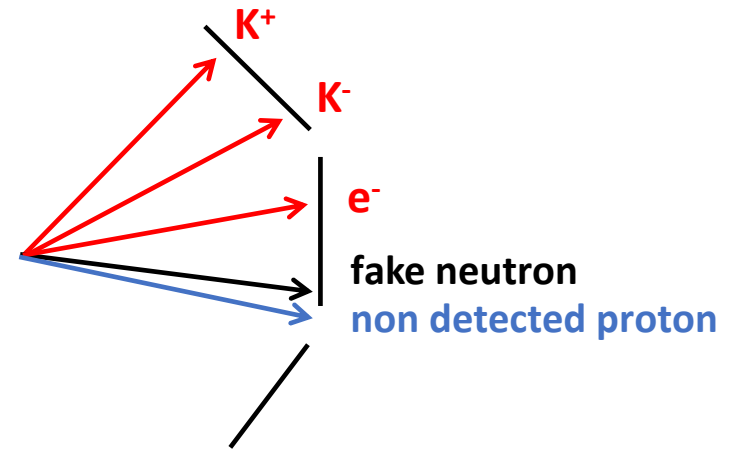
Neutron detection efficiency in the CND

Proton contamination

Missing_nucleon = the missing particle of this reaction : $e n \rightarrow e' K^+ K^- X$



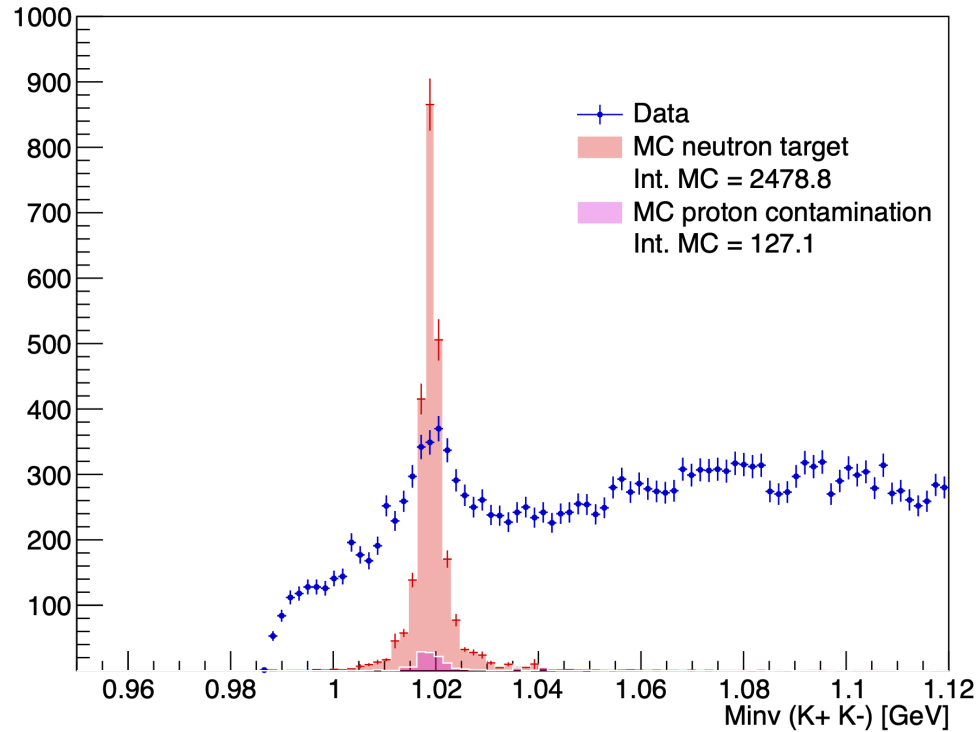
Good event with the angle between the missing nucleon and the neutron $< 5^\circ$



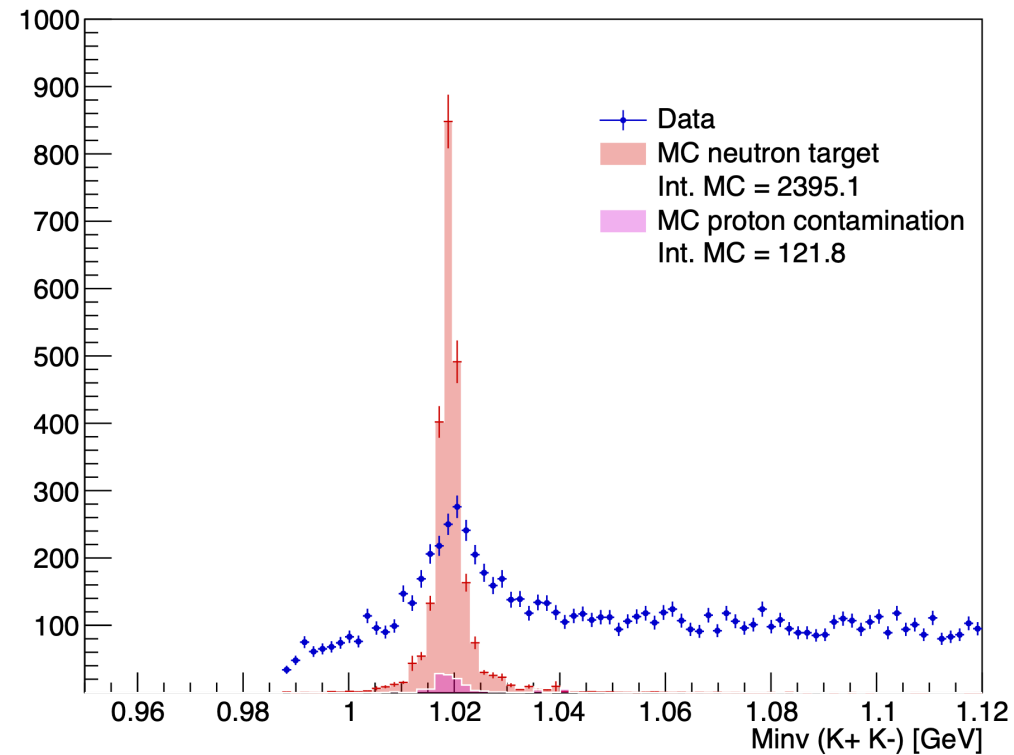
There is still a contamination when the **missing_nucleon** is a **non detected proton** and there is a **fake signal neutron** who pass the cut on the angle

→ Evaluate the contamination with simulation on **proton target** and apply the cuts of **neutron analysis** and see how many events pass the cuts

Influence of cuts on the invariant mass $K^+ K^-$:

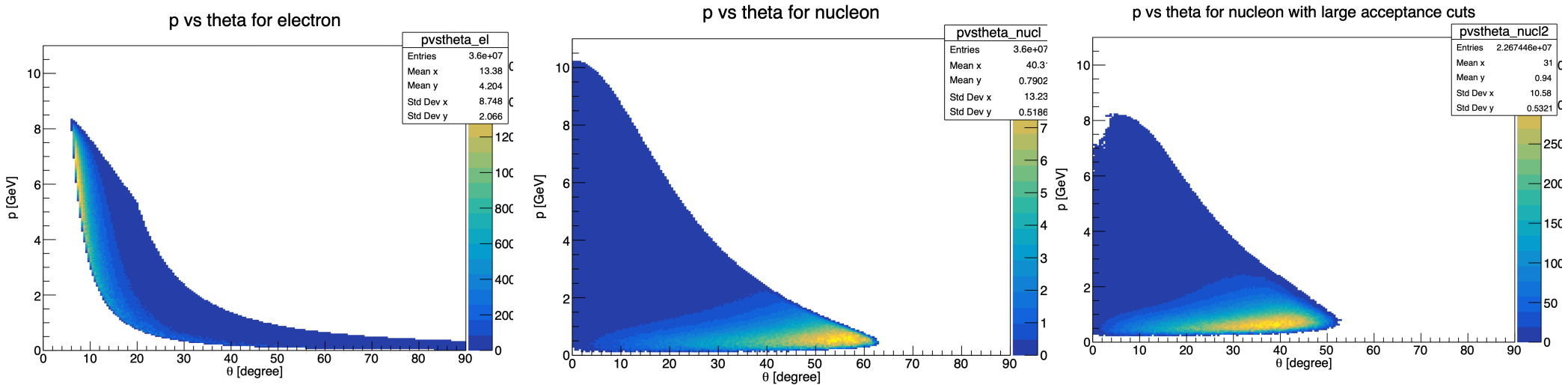


With angle cut

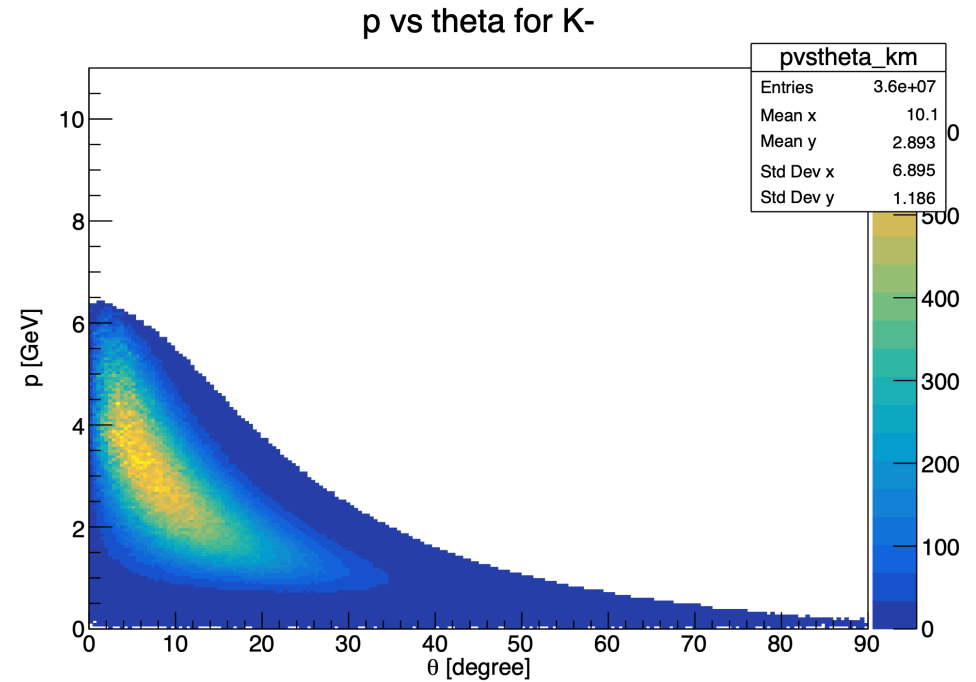
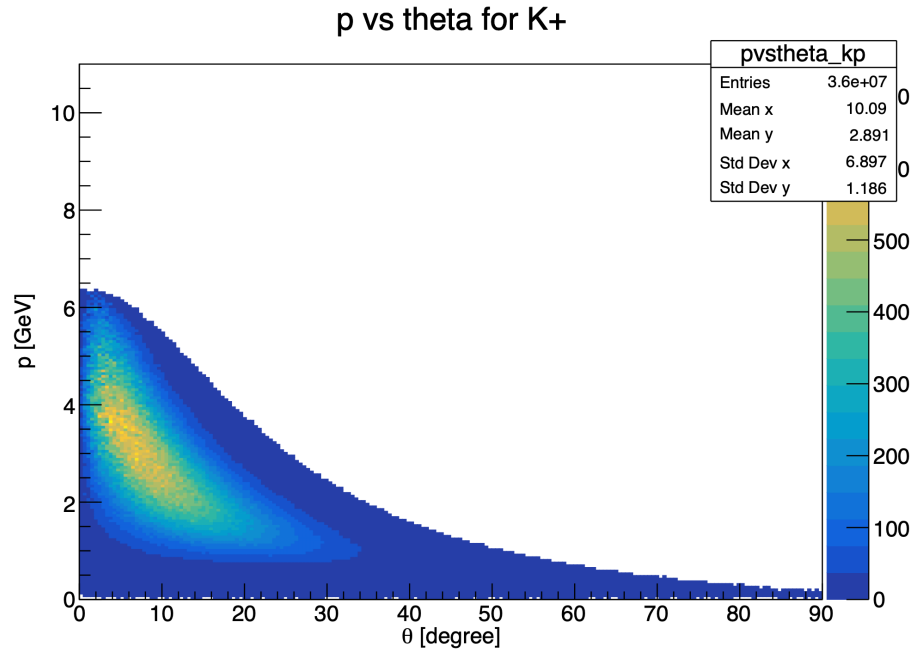


With angle cut and $K^+ K^-$ in FD

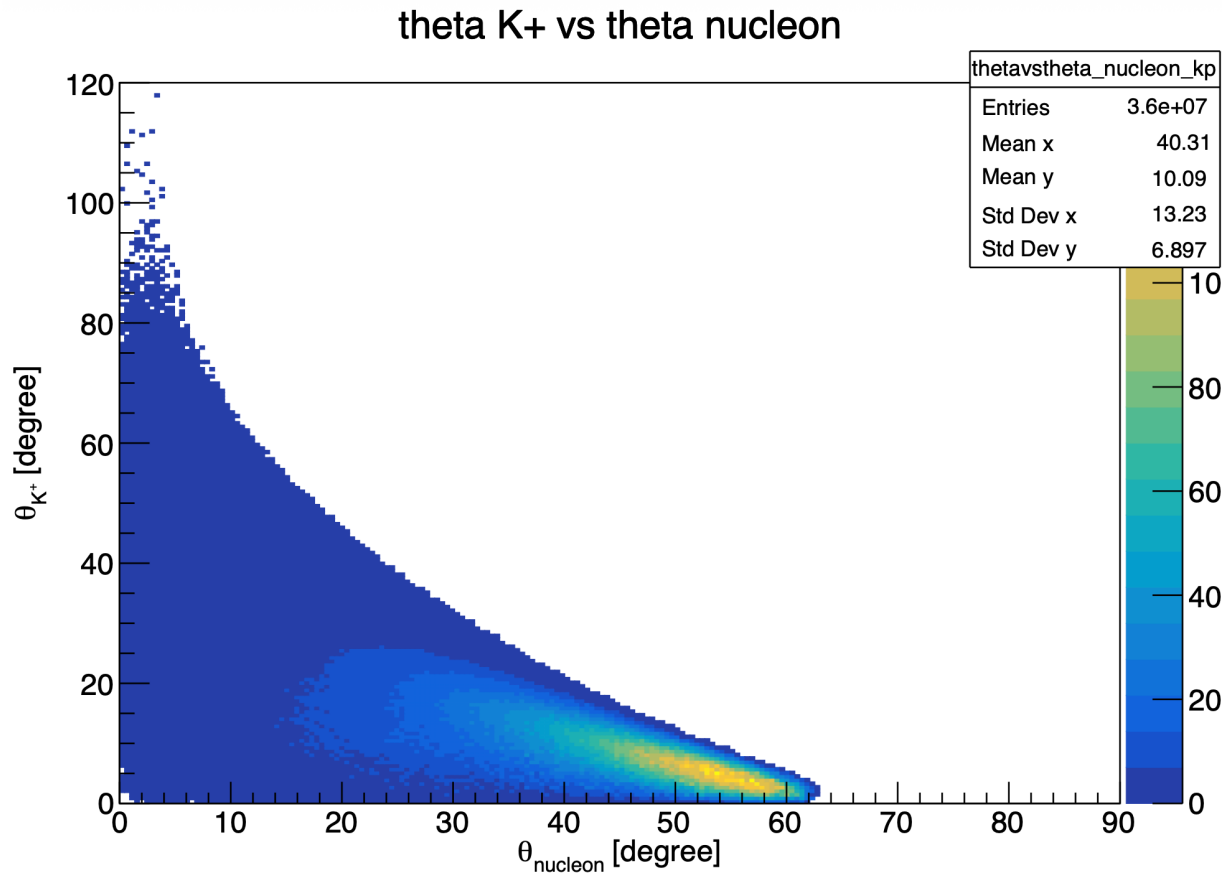
Generator plots



Generator plots

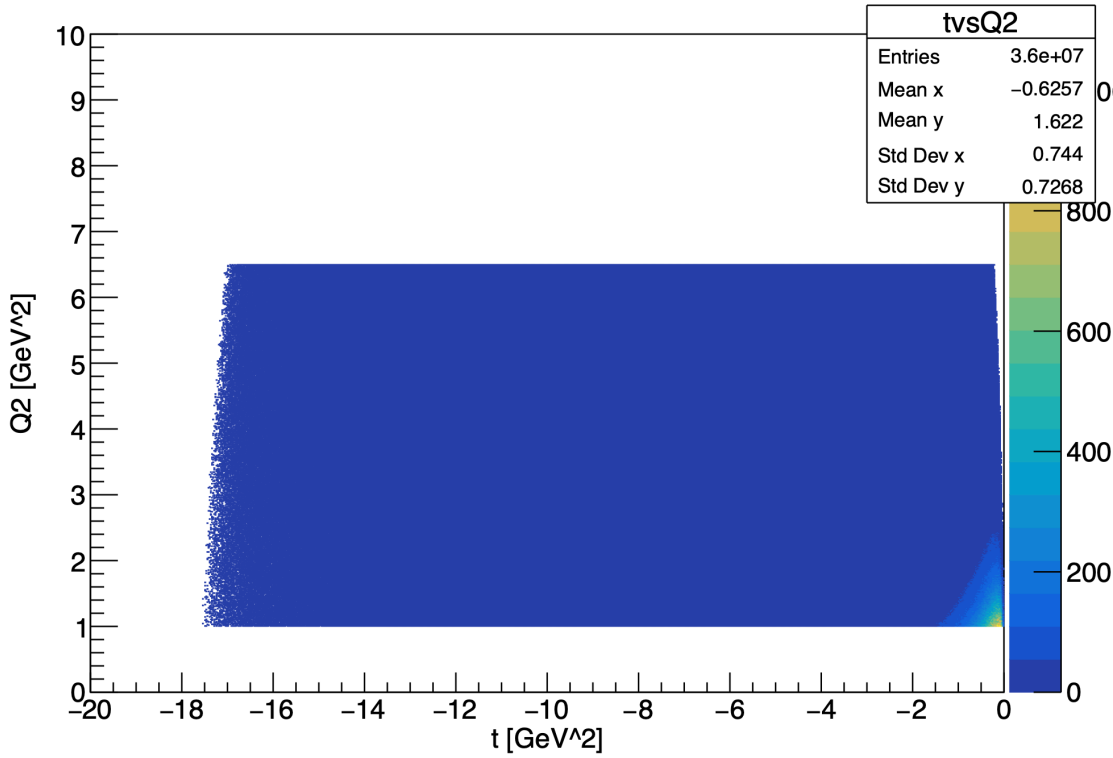


Generator plots

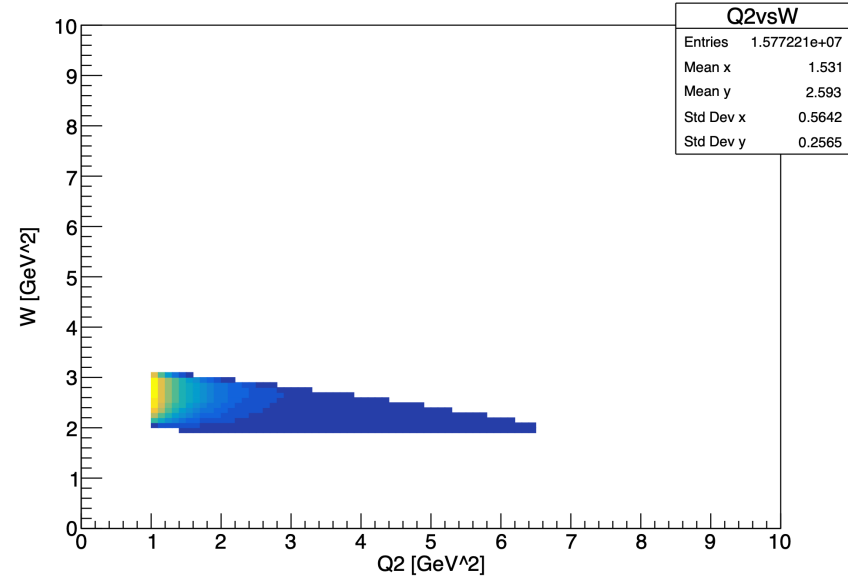


Generator plots

t vs Q2 with kinematics cuts

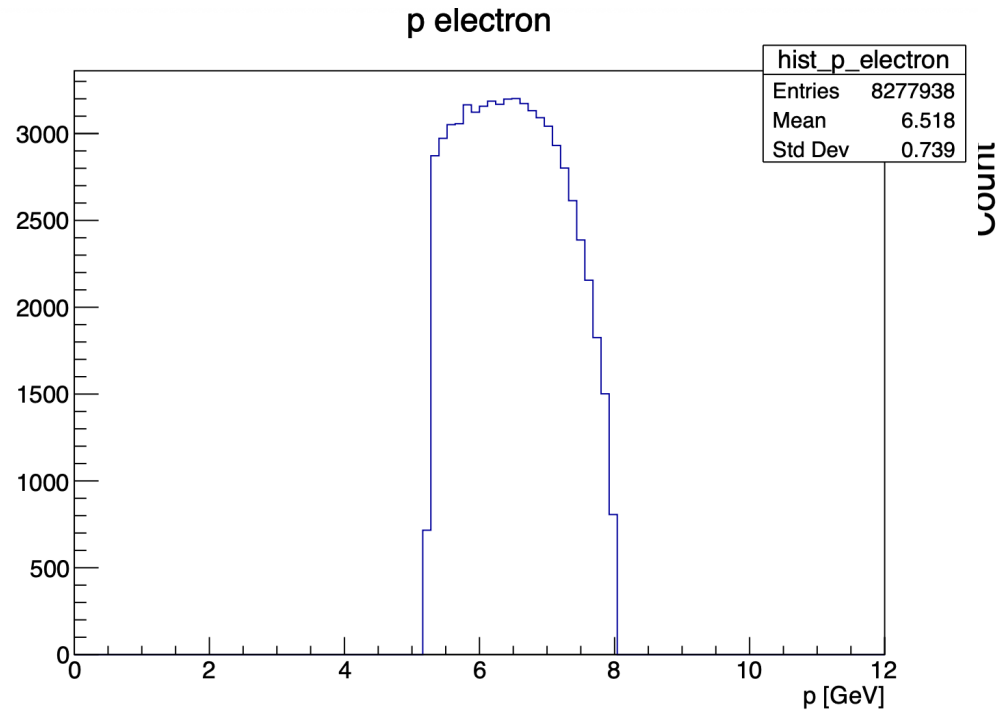


Q2 vs W

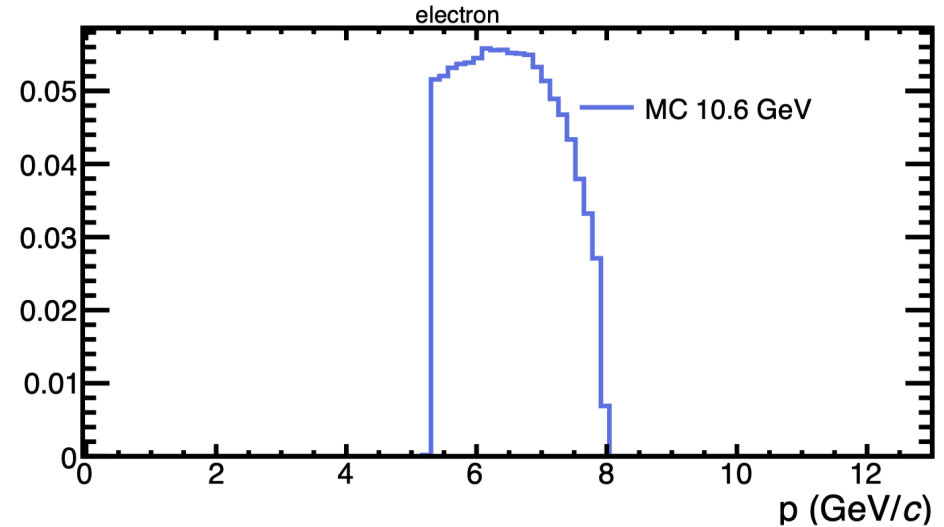


Generator plots

With this cuts : $5.25 < P_{\text{electron}} < 8 \text{ GeV}$ and $P_{\text{proton}} > 0.45 \text{ GeV}$



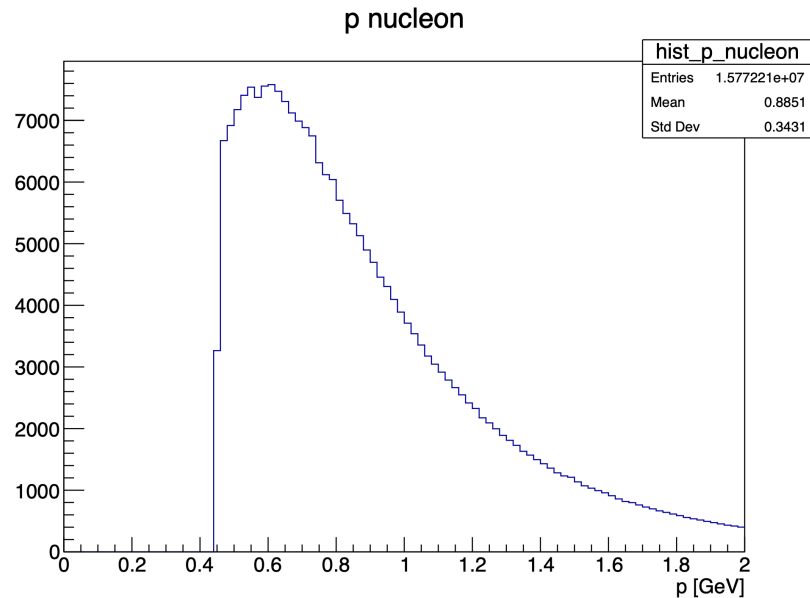
my generator



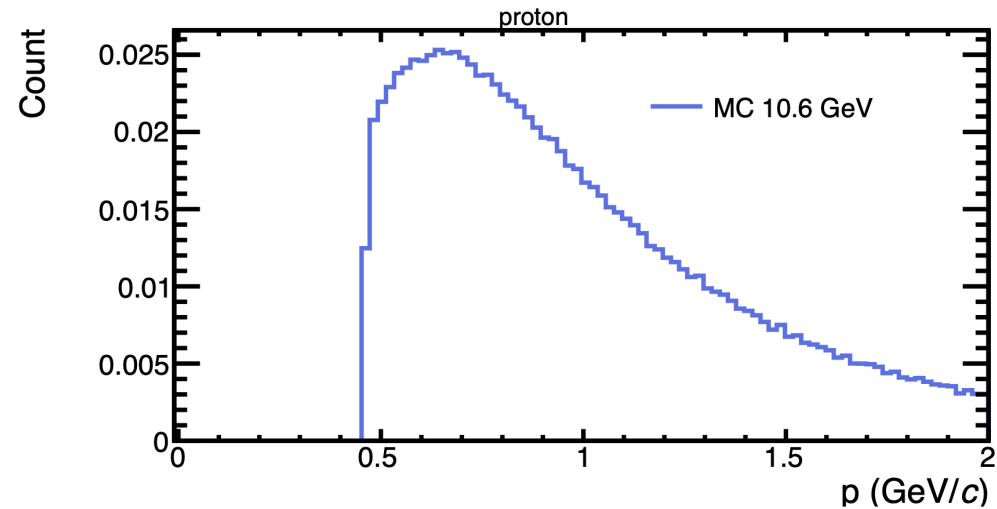
Bhawani plot

Generator plots

With this cuts : $5.25 < P_{\text{electron}} < 8 \text{ GeV}$ and $P_{\text{proton}} > 0.45 \text{ GeV}$



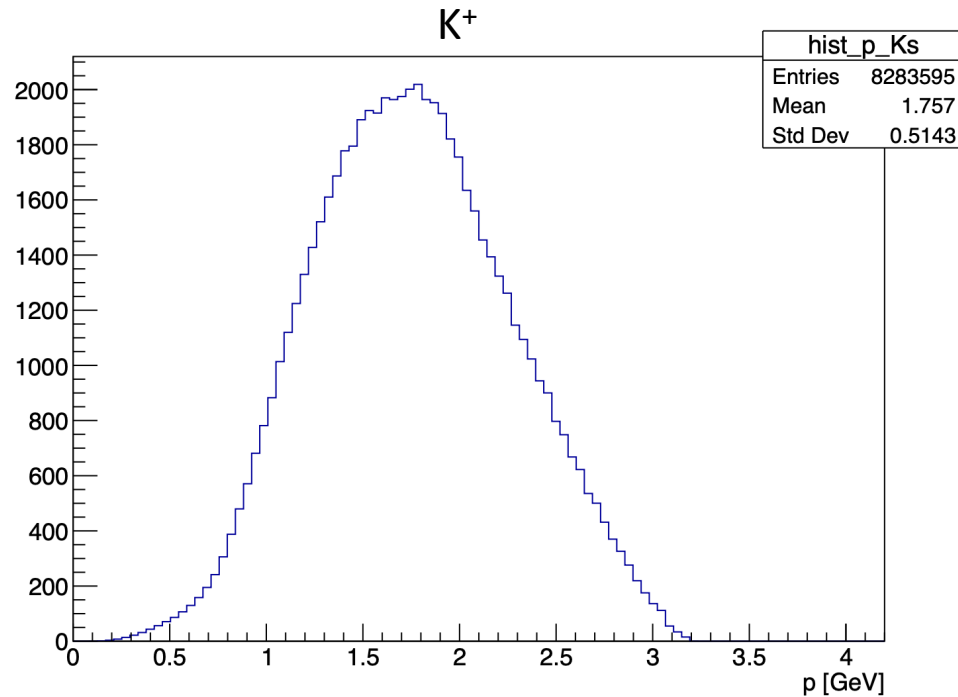
my generator



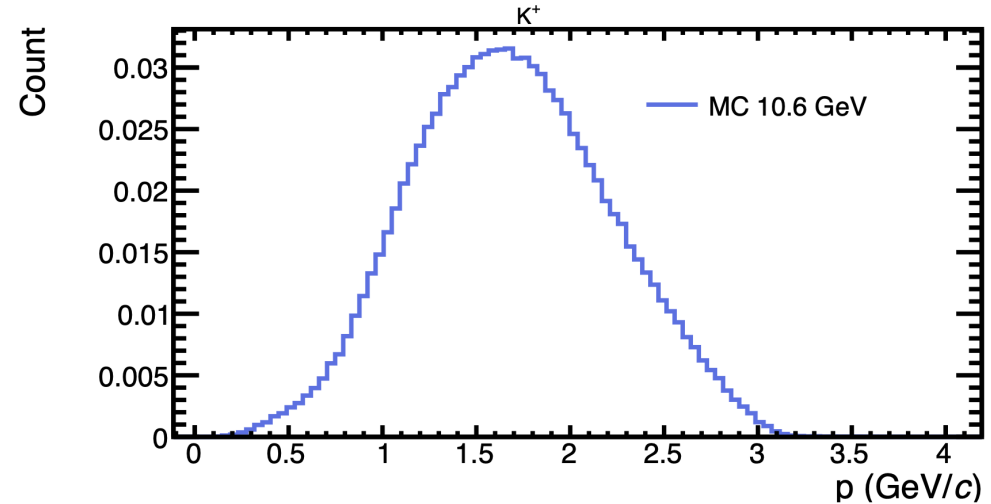
Bhawani plot

Generator plots

With this cuts : $5.25 < P_{\text{electron}} < 8 \text{ GeV}$ and $P_{\text{proton}} > 0.45 \text{ GeV}$



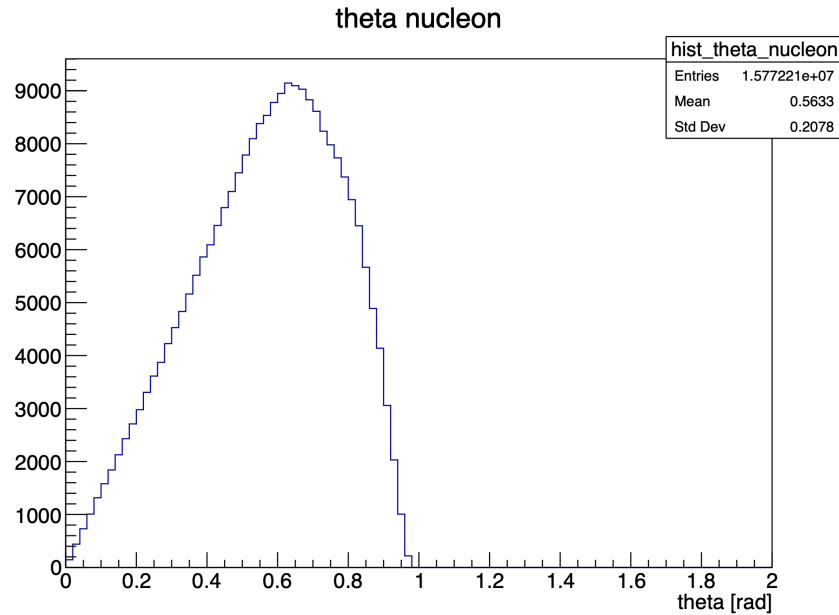
my generator



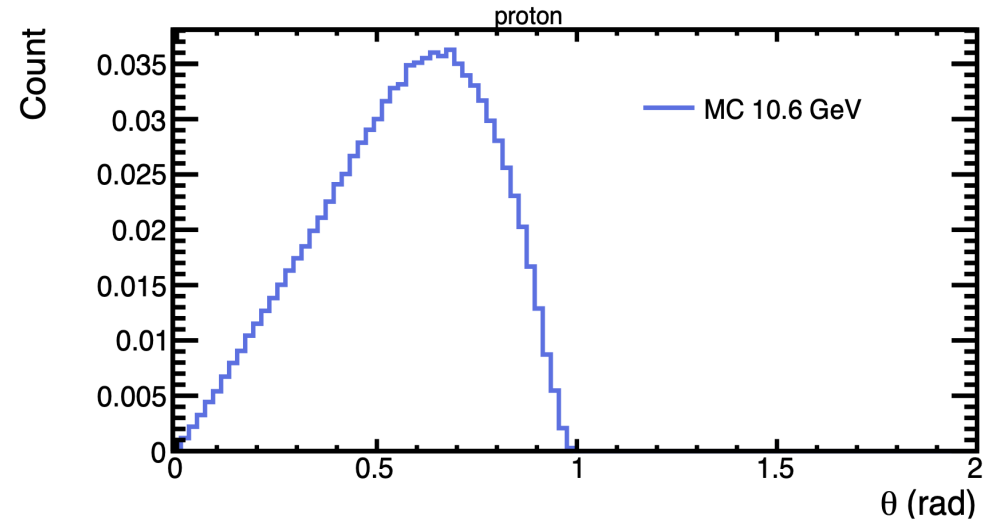
Bhawani plot

Generator plots

With this cuts : $5.25 < P_{\text{electron}} < 8 \text{ GeV}$ and $P_{\text{proton}} > 0.45 \text{ GeV}$



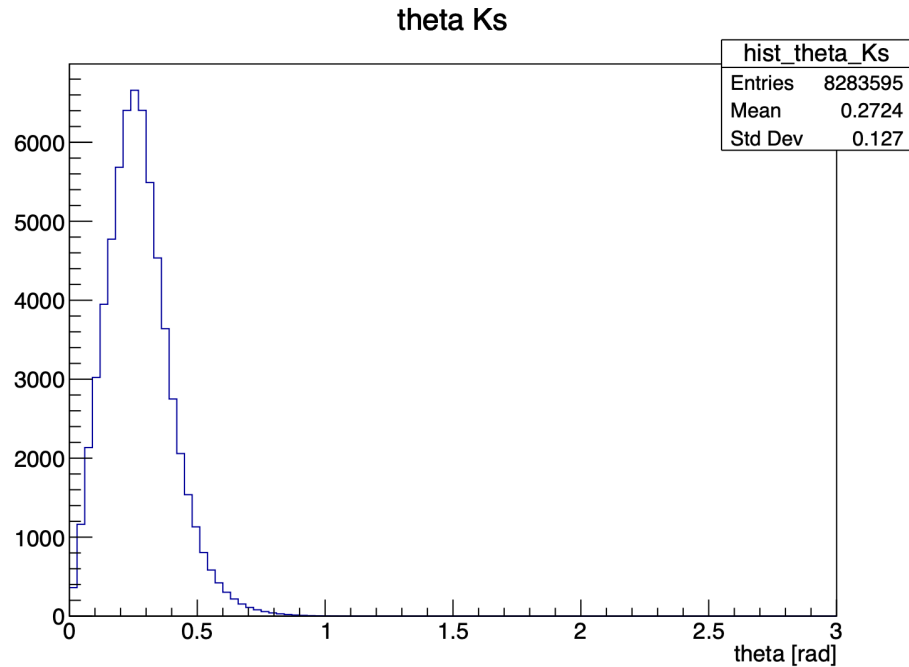
my generator



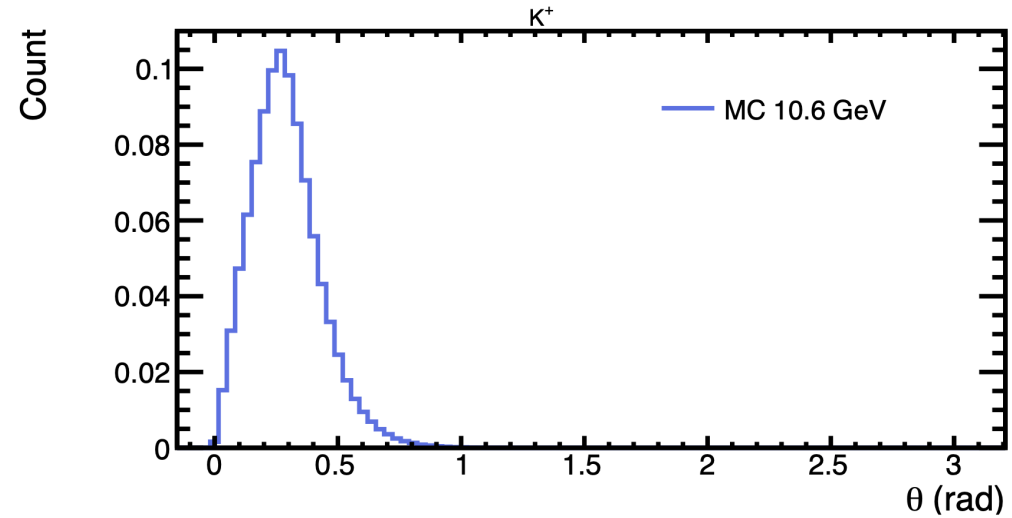
Bhawani plot

Generator plots

With this cuts : $5.25 < P_{\text{electron}} < 8 \text{ GeV}$ and $P_{\text{proton}} > 0.45 \text{ GeV}$



my generator

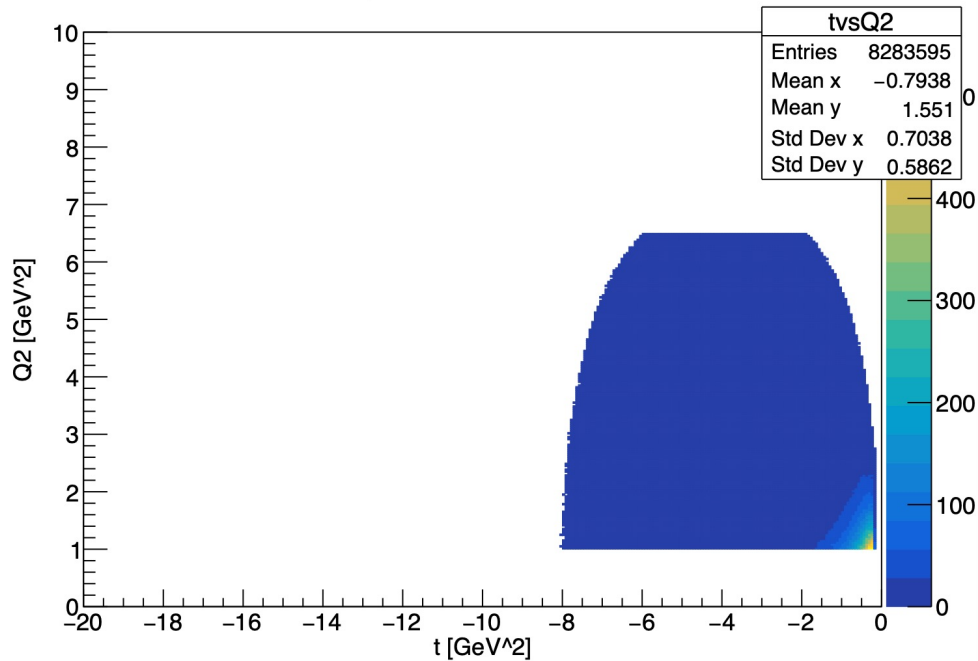


Bhawani plot

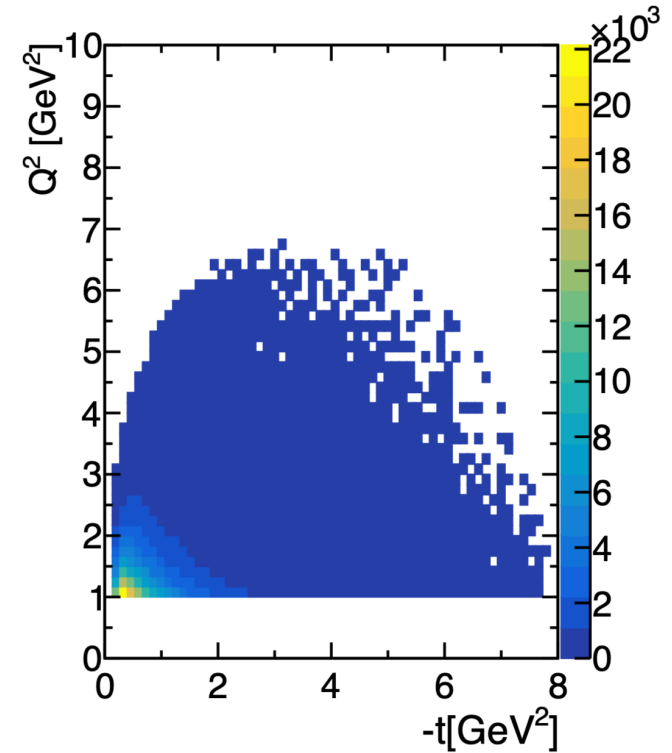
Generator plots

With this cuts : $5.25 < P_{\text{electron}} < 8 \text{ GeV}$ and $P_{\text{proton}} > 0.45 \text{ GeV}$

t vs Q2 with kinematics cuts

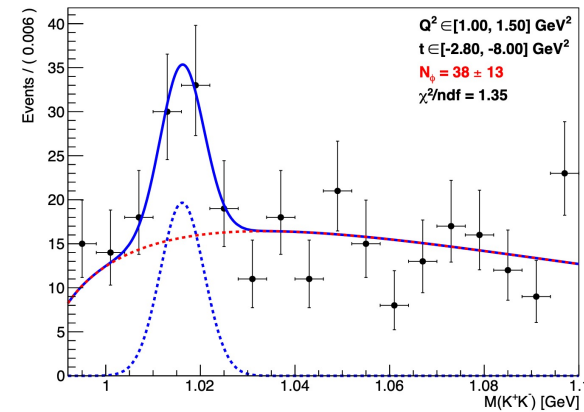
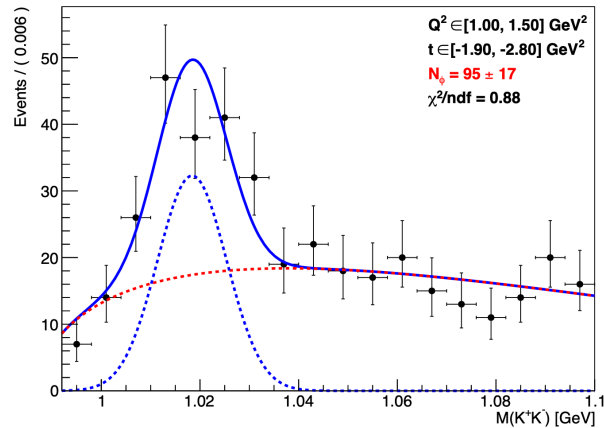
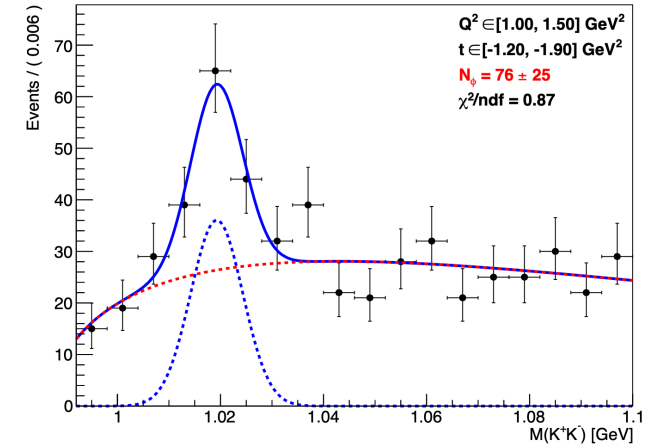
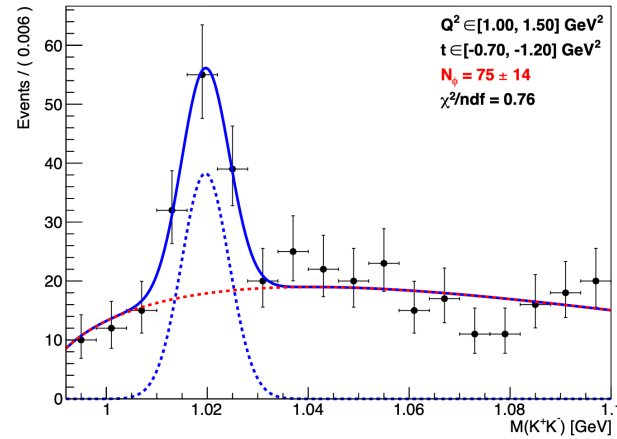
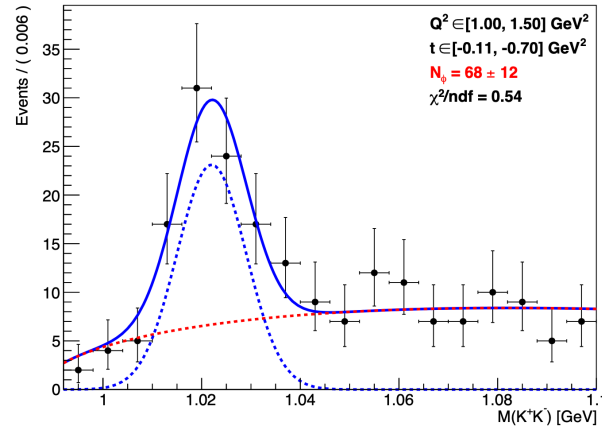


my generator

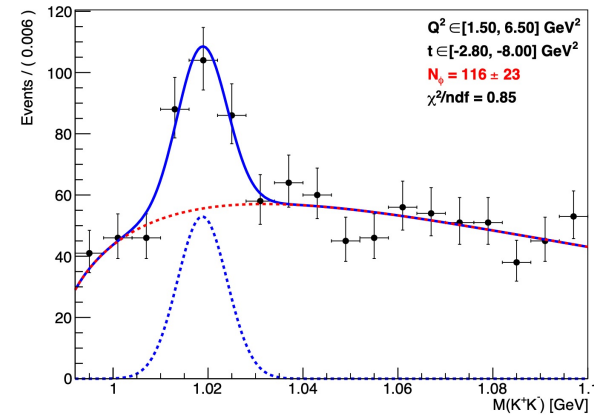
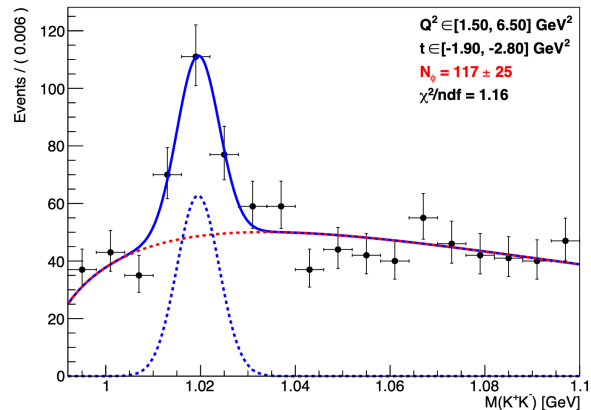
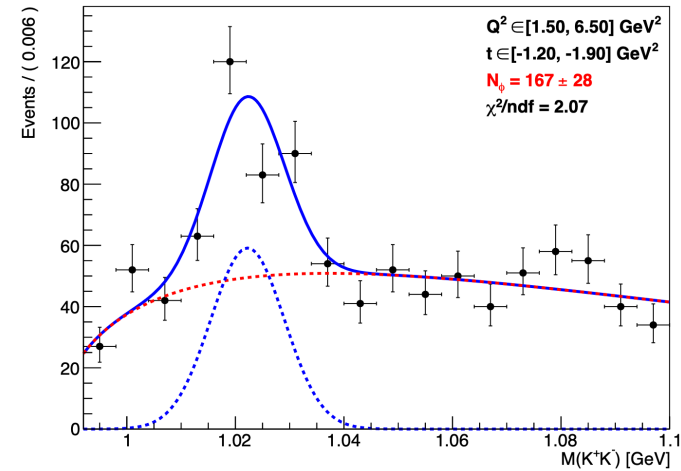
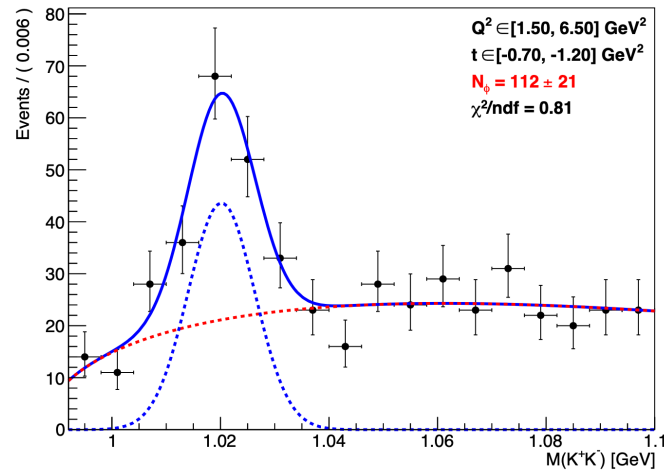
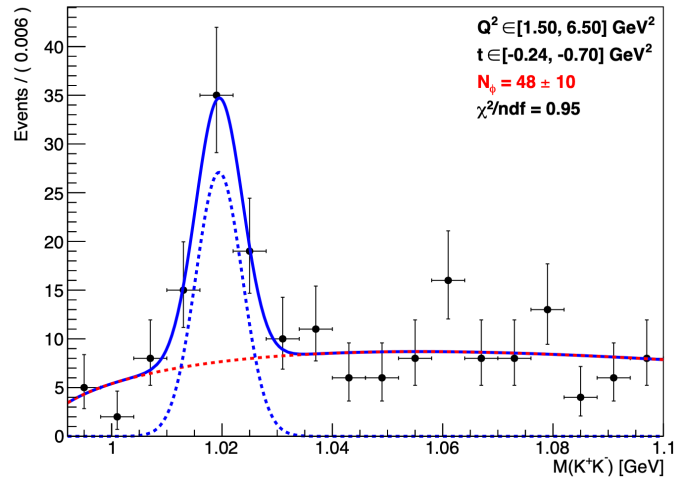


Bhawani plot

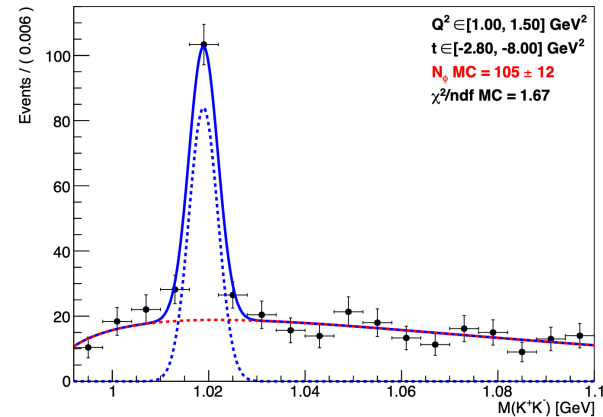
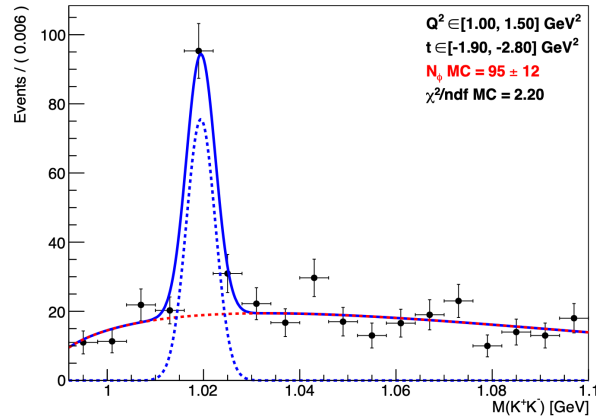
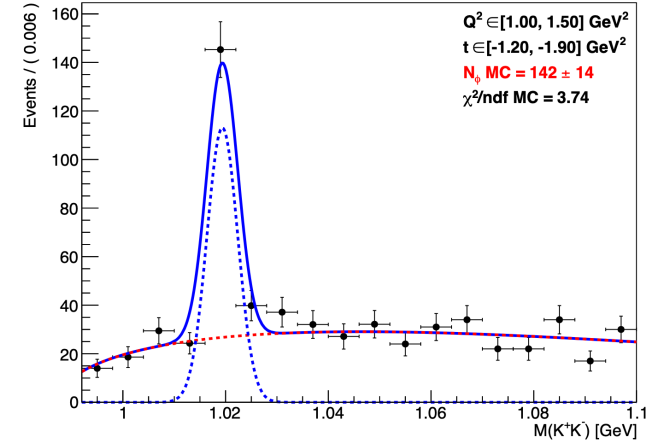
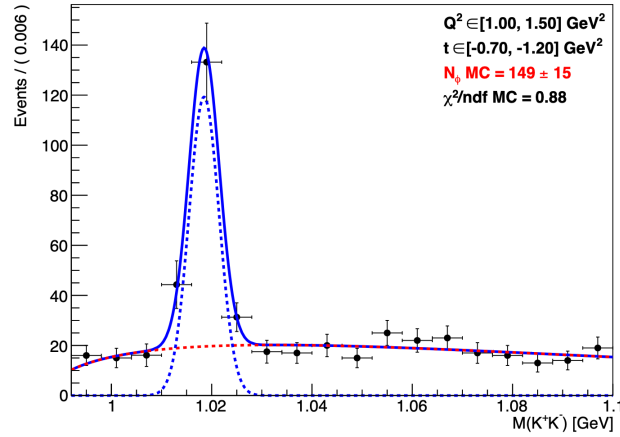
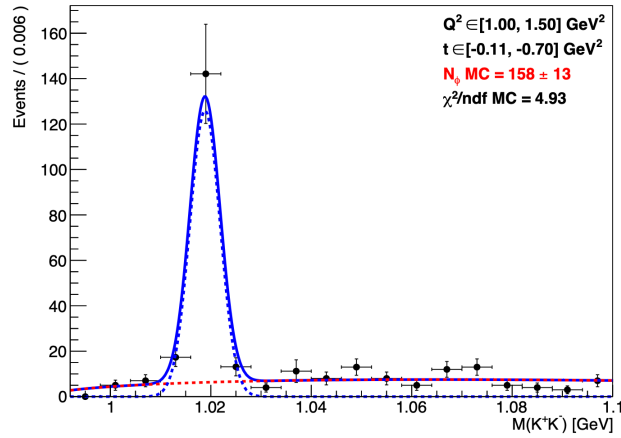
All fit data bin1 Q^2



All fit data bin2 Q^2



All fit mc bin1 Q^2



All fit mc bin2 Q²

